• Both pieces of work are OPEN BOOK. • Practical exam: – 90 minute Linux network and Linux configuration and troubleshooting. – you need to make specified changes to a Linux installation. – in Linuxzoo – 50% of module mark • Written exam: – 90 minutes – 20 short-answer questions in moodle – 50% of module mark

Commands

#prints a manual for the command entered
$man nameofcommand

#lists files and directories in a directory,list
$ls

#-a includes names begining with a dot
$ls -a

#-l long listing(includes permisions(if it starts with d its a direcotry or if its a - its a file), owner,group size of data, date and time created(or updated?) and name of file/directory
$ls -l

#calander, prints current month
$cal
#-s prints current month starting on sunday
$cal -s

#prints current working directory
$pwd

#create new direcotry called newdir
$mkdir newdir
#can make multiple directorys at once
$mkdir dir1 dir2 dir3
#can make direcorties inside a directtory without cd'ing into it
$mkdir dir1/dir4

#remove directory called netdir,only works if its empty
$rmdir newdir

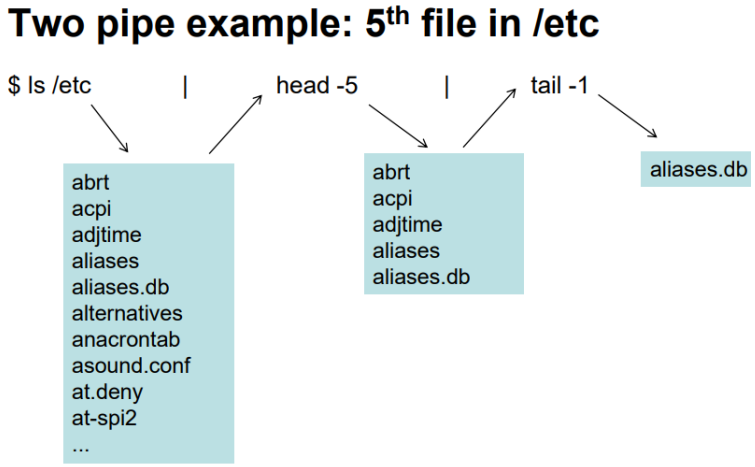
#removes direcotry called newdir and everything inside it even if its not empty,r is recursive and f is force, deletes everything from top to bottom of a direcotry tree without promting"are you sure"
$rm -rf newdir

#used to nagivate directories, cd /etc navigates into /etc , ../ takes you back a direcotriy, ../../ can take you back multiples.
#cd on its own takes you to the home directory

```
$cd /etc/
$cd ../

#duplicates a file, file1 is file being copied, file2 is new file being created(copy file1 to file2)
$cp file1 file2

#renames a file(esentially mv file1 into file2(creates file 2 and deltes file 1)
$mv file1 file2

#prints the contents of a file(concatenate inputs and prints them out)
$cat file1
hello
$cat file1 file2
hello
You

#user friendly cat, stops after each full page and says --more--
$more file1

#user friendly more, lets you move backwords
$less file1

#prints first “n” lines of a file (10 by default),can be changed by using -number
$head -2 file10
This is line 1
This is line 2
$head -3 file10
This is line 1
This is line 2
This is line 3

#prints last “n” lines of afile
$Tail -2 file10
This is line 99
This is line 100

#wc stands for “word count”, default tells you number of lines,words and characters
#can use -l -c or -w to see just number of lines,characters, or words.
$Wc /etc/passwd
46 77 2332 /etc/passwd

#redirect to a file, saves what would normally be printed to screen to file
#puts the outputs of the ls into outputfile
$ls -l /etc/hosts >outputfile
#redirection and pipes
> overwites whatever was in the file
>> appends to end of the file
< input comes from a file
```

| is a pipe used to redirect output into another command

#can pass output of one command into the input of the next command all in one line
$ls /etc|head -1
#ls's /etc and then only shows the first line of the output



Permissions
Each file ahas a owner, often each file also has one or more groups which it belongs to, linux feiles by default are considered to have 1 owner and 1 group

-rw-r--r--. 1 caine users 1025 Jun 3 2011 file1
File owner:caine - file has lkikely been created by user "caine" , file group: users - user caine likely belongs to group users.

Permissions
File or directory has various permissions and ownerships applied to it
Three file permissions:-r read permission, -w write permission, -x execute permission
Three permissions levels: -u user(creator of the object), -g group(a group identifier) -o other(everyone not in the user or group specified)
ls –l /etc/passwd
-rw-r--r--. 1 root root 1639 Sep 14 14:38 /etc/passwd
Owned by root, with group root. 1639 bytes in size, created on sep 14th at 14:30,1 link rw by user root, r by group root, r by other
First character - means normal file, d means direcotry, c means a character device(mouse ,keyboard, b means block device(die disk,scsi disk)

Ls -ld /home
wxr-xr-x. 2 root root 4096 Jul 27 13:38 /home
/home is a directory, owned by root in group root, UID root can do anythibng, group root can rx otrhers can rx, size is not really useful for directories

Finds a users id number
#id -u bob
#Bob UserID 1002
id -u jim

#jim UserID 1002

Chmod
Allows permissions of a file to be controlled, stands for “CHange the MODe” of a file…
Parameter 1 is the change requested. Parameter 2 is the file or directory name being changed.
The change is one or more of “ugo”, then “+-=”, then zero or more of rwx - + adds permissions to the current set, - takes permission away, = does not care what current permissions are, and deltes them before setrting the perimssions you have specified.

Numeric Notation
An older way of looking at permissions, still needed for some commands and a fast way of changing multiple permissions, based on octal, 4 digits long. Digit 0 is usually -, 1 is owner, 2 group,3 other,

Octal	Binary	Perms	Octal	Binary	Perms
7	111	rwx	3	011	-wx
6	110	rw-	2	010	-w-
5	101	r-x	1	001	--x
4	100	r--	0	000	---

If User rwx, Group rx, Other rx,
– Symbolic –rwxr-xr-x
– Numeric 0755
If User rwx, Group x, Other none
– Symbolic –rwx--x---
– Numeric 0710

Executable Scripts
Executable scripts are programs which contain commands,these can be executed at hte command prompt, if you want to create a file to contain everything from a webpage:select what you want and then “cut”, use putty to ssh or telenet to your machine, if say, your script is called demo go to the right direcotry and do cat > demo paste by pressing right mouse button, once it is all pasted in, press cntrl-d
Now you need to make the script execuatable
Now you can run the script

File Wildcard Expansions
When a command takes a filename parameter you can specify the exact name, but sometimes it is more useful to specify a “wildcard” which allows you to specify a range of filenames. For example you might want to: copy all files which end in .doc,delete all files which end .deleteme, see how many lines in total are in all the files “evidence01.dat to evidence99.dat

When you use this to ddescribe filenames, it is called filename expansion, or globbing, or filename wildcardds. Do not confuse this with regular expressions as they are not interchangable.

Wildcards
Parameters which match filenames dont have to be complete. You can pattern match with the characters “?” for a single character and “*” for a number of characters.
$ls
aaa aab abb
$ls aa?
aaa aab
$ls a*
aaa aab abb

Wildccard[set]
You can pattern match with a set of characters. For instance, you want files which end with a or b
$ls
aaa aab aac zzb zzc
$ ls aa[ab]
aaa aab
$ls *[ab]
aaa aab zzb

Regular expression
Regular expression is a standard way of defining pattern matches. It has a number of versions, but in each the core syntax is the same. Used in a variety of search commands in linux, very different from filename expansaions used in terminal commands like ls and cp. One command which can use regexp is grep - grep searches for pattern in the contents of a file and reports the matches. To trigger grep to use the regexp discussed here you muist use the option “-E”.
Does the file “file1” have the string “hello” in it: $ grep –E ‘hello’ file1

Regexp: Single Characters
Normal characters like a or z look for those characters
Some other characters have a special meaning. – A dot “.” character can match any character. – [...] – Characters within square brackets mean match 1 character and that character must be one of those shown in the square brackets – \. (slash dot) – To actually look for a dot and not represent any character – \[(slash bracket) – Actually look for a square bracket
In effect if a character has a special meaning you can stop the special meaning by putting a \ character in front of it - this is called an escape sequence.
$ grep –E ‘teleplastic’ /usr/share/dict/words teleplastic $ grep –E ‘teleplasmic’ /usr/share/dict/words teleplasmic $ grep –E ‘teleplas.ic’ /usr/share/dict/words teleplastic teleplasmic
$ grep –E ‘publicise’ /usr/share/dict/words publicise $ grep –E ‘publicize’ /usr/share/dict/words publicize $ grep –E ‘publici[sz]e’ /usr/share/dict/words publicize publicise

$ grep –E ‘etc\.’ /usr/share/dict/words etc. $ grep –E ‘etc.’ /usr/share/dict/words etc. etch
By default regexp looks for a match somewhere on each whole line. $ grep –E ‘bump’ /usr/share/dict/words bump bumpy unbump unbumpy
If you want to say “match from start of line” you say “^” (hat) at the beginning of the regexp. • If you want to say “match to the end of line” you say “$” (dollar) at the end of the regexp
$ grep –E ‘^bump’ /usr/share/dict/words bump bumpy $ grep –E ‘bump$’ /usr/share/dict/words bump unbump
$ grep –E ‘^bump$’ /usr/share/dict/words bump

Repetition
You can add special characters to say how many times the previous character should appear
c* character star - 0 or more “c” characters
C+ charactrer plus - 1 or more c characters
C? Character questionmark - 0 or 1 of the c character

$ grep –E ‘^a.*z$’ /usr/share/dict/words abuzz allez • Word has ‘a’ then ‘b’ then ‘c’, with 0 or more characters in between. $ grep –E ‘a.*b.*c’ /usr/share/dict/words diabetic

The regular expression engine
Greedy wildcards • Wildcards match as much as possible, then try less and less until they work.
So “^a.*b.*c$” matching “amebic” is:

a	.*	b	.*	c
a	mebic	-	-	-
a	mebi	-	-	-
a	meb	-	-	-
a	me	b	ic	-
a	me	b	i	c

So the first one matches 5, then 4, etc, until something goes right. This retry process is called “backtracking”

Backreferences
Sometimes you want to group part of a regular epression statement, and refuse what that matched in a later part of a expression, for example, look for a 3 character string beginning with a which occurs twice in aa line This could match WALL-TO-WALL and RAZZMATAZZ
To do this we need to group the first match, then reuse the group with a backreference. • A group is part of a match surrounded by brackets “(....)”. – The brackets are not treated as something to look for, but are special characters. • The point where you want to say “the thing which was in the brackets” you say “\1”, where 1 is the bracket number (e.g. You can have more than 1 set of brackets).
grep –E ‘(a..).*\1’ /usr/share/dict/words abracadabra
In other words: – Search for “a..” (three characters where the first character is A) – Remember that match and call it GROUP 1. – Then 0 or more characters are matched – Then the same three character combination called GROUP 1 has to appear

Look for words in the dictionary which have three vowels appearing together, then the SAME three vowels appearing together AGAIN in the same word.
grep -E '([aeiou][aeiou][aeiou]).*\1' /usr/share/dict/words
aeonicaeonist Andreaeaceae homoiousious

Command Summary: – md5sum – cmp – sha512sum – grep – find – file
Top of the tree is “/”, pronounced “slash” or “root”. All files and directories hang off this • Off this are directories like /etc and /home • Off /home is a directory “caine”. • So two levels above /home/caine is / • /home/caine is caine’s HOME directory.

File
In windows the file extensioun says what a file is - gordon.doc,this is a word document, due to a file association(.doc->word)
Secretive windows users may change an extension to hide evidence.
In linux, there are no file extensions, and thus all associations are calcualted from the contents of a file - often called signature analysis, in linux there is a useful tool for this analysis - the command is file

Examples $ file /bin/ls (the ls command) /bin/ls: ELF ... Executable...dynamically linked ... $ file randomfile (a jpeg image with random name) randomfile: JPEG image data, JFIF standard 1.01 $ file /etc/hosts (just plan text about system hostnames) /etc/hosts: ASCII text $ file privateimg (a GIF with a silly name) privateimg: : GIF image data, version 89a, 627 x 671

Hashing
If a file is copied and renamed, how can we know both files are the same, one way is to hash all the files, then see if the hash numbers are identical. A hash is an algorithm which reduces a large file into a simple short number, in a way that two files which are identical has the same hash, but two different files should have different hash numbers.

Simple hash - sum mod 8
Consider a hashing algorithm which adds all the bytes of a file together then MODs the total by 8 - MOD 8 is the remainder of a divsion by 8

File 1	File 2
5	1
6	2
1	7
3	1
(5+6+1+3) => 15 15 / 8 => 1 remainder 7	(1+2+7+1) => 11 11 / 8 => 1 remainder 3

So the hash of a file1 is 78 and the hash of file 2 is 3 they are different hashes thus different files. This is a stupid hash algorithm as there are many files which will have the same hash, but which are in fact different.

Md5sum
Calculates an 128 bit md5 checksum - takes 1 parameter: the file being analysed
$ ls file1 file2
$ md5sum file1 817ea56a11b3f9b476e0940f353c782a file1

$ md5sum file2 817ea56a11b3f9b476e0940f353c782a file2

Hash Collisions
If two files have different hash values then they are not identical.
If two files have the same hash values then they are probably identifical
If two files are different but have the same hash they are referred to as a ahash collision or a false positivve. There are many possible files which will return the same hash, the better the hash function the less chance of a hash collision.
The cmp command does a binary check - if “cmp” prints anything the files do not match
If cmp prints nothing they are identical
$ cmp file1 file2
file1 file2 differ: byte10, line 1

Sha512sum
Calculates an 512bit sha checksum takes 1 parameter: the file being analysed

Sha collisions
Sha 512 has no known hash collisions, it is therefore almost certain that if two files have the same SHA 512 hash then they are identical, does not do any harm to check with cmp
But sha 512 hashes are much much bigger than md5 128bit hashes if you have to write them down it may be tiring and error prone

Find
Find command is very powerful at searching for filenames. If you know something about the files you are looking for, find can locate all files in a tree which match the conditions
It has slighjtly complex parameter format: - parameter 1: the top of the tree you want to search in, the remaining parameters are either - tests which have to be true before an action is carried out. Different tests are ANDed together by default. Or actions which are carried out when all the rules are true.
When find locates a matching file it carries out one of more actions - for our studies we will only print to the screen, or execc a command, print is teh default action, so in our case we will not need to specify any actions. Possible actions are things like -print,-exec,-delete and many more
Where rules have a numberical parameter, the number can be -n test to see if the nunmber is N, +n test to see if a file has a number greater than N, -n test to see if a file has a number less than N
Basic rules include: -atime N file accessed N*24 hours ago, -atime+1 looks for files accessed >1 day ago, atime 1 looks for files accessed in the last 24 hours.
-user USER files owned by a particular USER. -group GROUP files owned by a particular GROUP. -name NAME files named name, can use filename wildcards. -prem MODE files with MODE chmod permissions. -size N files are size N., each the number with “c” for size in bytes. -type C C can be “d”(directory), f(file) pus others

$ cd /home/caine $ ls -l drwxrwxr-x. 2 gordon gordon 4096 Jan 30 11:52 dir1 drwxrwxr-x. 2 gordon gordon 4096 Jan 30 11:52 dir2 -rw-rw-r--. 1 root caine 187 Jan 30 11:51 file1 -rw-r--r--. 1 gordon caine 157 Jan 31 16:40 file2 $ ls -l dir1 -rw-r--r--. 1 root gordon 187 Jan 30 11:51 file3 -rw-rw-r--. 1 gordon gordon 147 Jan 31 16:40 file4 $ find /home/caine –size 187c /home/caine/file1 /home/caine/dir1/file3

$ cd /home/caine $ ls -l drwxrwxr-x. 2 gordon gordon 4096 Jan 30 11:52 dir1 drwxrwxr-x. 2 gordon gordon 4096 Jan 30 11:52 dir2 -rw-rw-r--. 1 root caine 187 Jan 30 11:51 file1 -rw-r--r--. 1 gordon caine 157 Jan 31 16:40 file2 $ ls -l dir1 -rw-r--r--. 1 root gordon 187 Jan 30 11:51 file3 -rw-rw-r--. 1 gordon gordon 147 Jan 31 16:40 file4 $ find . –user root ./file1 ./dir1/file3
$ cd /home/caine $ ls -l drwxrwxr-x. 2 gordon gordon 4096 Jan 30 11:52 dir1 drwxrwxr-x. 2 gordon gordon 4096 Jan 30 11:52 dir2 -rw-rw-r--. 1 root caine 187 Jan 30 11:51 file1 -rw-r--r--. 1 gordon caine 157 Jan 31 16:40 file2 $ ls -l dir1 -rw-r--r--. 1 root gordon 187 Jan 30 11:51 file3 -rw-rw-r--. 1 gordon gordon 147 Jan 31 16:40 file4 $ find . –name ‘*[23]*’ ./dir2 ./file2 ./dir1/file3

“-exec”
Instead of “print” you may want the action to carry out a command every time there is a match - for instance, “ls -l” the filenames which end in “conf” in /etc
At the end of the find line add -exec COMMAND … {} … \;
Command is the command you want to run, you can include ther options in this section, where you want the name of hte file found to appear in the exec command write the open and close curly brackets “{}”. End the exec area with slash and semicolon “\;”

$ find /etc -name 'c*.conf' -exec ls -l {} \; -rw-r--r--. 1 root root 950 May 30 2011 /etc/sysconfig/cgred.conf -rw-r--r--. 1 root root 91 Jun 2 2011 /etc/gdm/custom.conf -rw-r- • Find all files in /home/caine which end “.del” and delete those files. $ find /home/caine -name '*.del' -exec rm {} \;

Networked Services
Users, Permissions, Processes
Linux is a file based system. Everyone on linux is either a file or a directory. A file which is executed becomes a process. Processes can be seen as files too. Devices, such as scanners and hard drives are also files.

Directories
/bin:this contains commands a user can run like ls but which might be needed during boot
/boot: things needed during the boot process.
/dev: contains devices like the mouse.
/home: this is where users store their files
/tmp: temporary storage for users and the system
/var:system files which can change
/etc: system config files which dont change
/lib and /lib64: where all the system libraries live
/proc : files which represent the running system(like processes)
/sbin:commands which only an administrator would want
/usr : commands whiuch are never needed during bootup

USERS
UID and GID
In unix, there are User IDs and Group IDs. User IDs uniquely identify a particular user. Group IDs allow users to be collected into groupings. Groups could be used to allow friends to share files, which stopping people not in that group of friends from reading the files

Users
User details are stored in 4 files. – /etc/passwd - General User details. – /etc/shadow - User passwords. – /etc/group - The users' groups. – /etc/gshadow - Passwords for groups

cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
bin:x:1:1:bin:/bin:/sbin/nologin
daemon:x:2:2:daemon:/sbin:/sbin/nologin
Username, x,UID,GID, text description(finger information), home directory, login shell. Field delimiter is a colon.

Head -3 /etc/shadow
root:1RcFIaOlb$bwl5dvTECg3M1ZgMQ7e6I.:12663:0:99999:7:::
bin:*:12621:0:99999:7:::
daemon:*:12621:0:99999:7:::
Passwords are md5 hashed. (with random seed). Means no password these "users" cant log in. Shadow passwords can expire and have rules. Passwords are hashed(with random salt) 6 indicates sha512 hash. * means no password - these "users" cant log in. shadow passwords can expire and have rules.

Tail -3 /etc/group
gdm:x:42:
dovecot:x:97:
Mysql:x:27:
Group contain group names,x,and the number which defines that group uniquely. After the last : can be a list of users who are in that gorup
Friends:x:500:gordon,andrew
Allows people to change groups on a password. Not often used, but when done the password is placed here where hte x is

EXTRACTING INFORMATION
How can we extract information from structured files?
You know how to use > for redirection and how to use grep, possibly with a pipe, to extract relevant rows of information. But that doesnt allow us to, for example, find all the usernames beginning with "a". We can use a regular expression, "^a", to extract the relevant rows from /etc/passwd.
$ grep "^a" /etc/passwd
adm:x:3:4:adm:/var/adm:/sbin/nologin
apache:x:48:48:Apache:/var/www:/sbin/nologin
andrew:x:501:500:Andrew Cumming:/home/andrew:/bin/bash

Cut
The grep got the information, but also lots of other pieces of info. How can we extract only the usernames? The command "cut" chop things out of a line. It will split a field out of a line so long as it knows what character is the delimiter(marks the end of one field and the start of another). In /etc/password, ":" is the delimiter, so cut is -d":"
The username is in field 1, so -f1
$ grep "^a" /etc/passwd | cut –d":" –f1 adm apache andrew

FILE OWNERSHIP and ADDITIONAL PERMISSIONS
Chown
Used to change owner, optionally also group
Chown <new owner>[.<new group>] <file>
> ls -l /tmp/test -rw-rwxrwx. 1 root root 0 Sep 23 15:47 /tmp/test > chown ftp.mem /tmp/test > ls -l /tmp/test -rw-rwxrwx. 1 ftp mem 0 Sep 23 15:47 /tmp/test

Chgrp
Used to change group only

> ls –ld /home
drwxr-xr-x. 2 root root 4096 Jul 27 13:38 /home
/home is a directory, owned by root in group root.
UID root can do anything, group root can rx, others can rx. Size is not really useful for directories. The "." immediately after the permissions indicates that alternative access methods exist. If this is a " "(space) there are no additional methods. "."(dot) indicates a selinux security context and "+"(plus) indicates a combination of access menthods. The "2" indicates that there are two hard links to the file.

Alternative Access Methods
ACL access methods allow you to set fine-grained permissions
> touch test > setfacl –m user:root:rwx test > ls -l test -rw-rw-r--+ 1 gordon gordon 0 Aug 30 15:25 test > getfacl test user::rwuser:root:rwx group::rwmask::rwother::r--
SELinux access methods map complex process rules to file context information, e.g. The web server can only see files in the "httpd_user_context_t" context.

Numeric Notation
An older way of looking at permissions. Still needed for some commands, and a fast way of changing multiple permissions. Based on octal, 4 digits long. Digit 0 is usually 0, 1 is owner, 2 group, 3 other.

Octal	Binary	Perms	Octal	Binary	Perms
7	111	rwx	3	011	–wx
6	110	rw–	2	010	–w–
5	101	r–x	1	001	––x
4	100	r––	0	000	–––

Default permissions: umask
Umask 022
When a command creates a file or direcotry the default is:
– rwxrwxrwx – for directories
– rw-rw-rw- – for files
The value of your umask is subtracted from the numeric protection code.
So removing write for group and other you need to know that 2 stands for w, and this for:

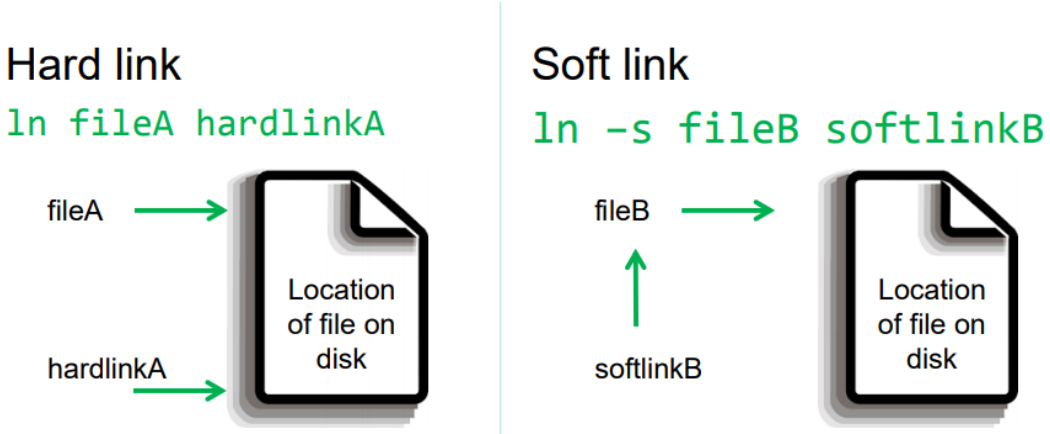
– rw-r--r-- (644) - Write only for owner. – This is numerically, 666-022 => 644 – So the umask is 022.

The umask mask
0022
– Col 0 is always 0, Col 1 is OWNER – Col 2 is GROUP, Col 3 is OTHER

Octal	Binary	Perms	Octal	Binary	Perms
0	000	rwx	4	100	–wx
1	001	rw–	5	101	–w–
2	010	r–x	6	110	––x
3	011	r––	7	111	–––

Hard and Soft Links
File links
Sometimes we want to have hte same file contents in two or more different places-- edit one version and you edit all versions, you only use up the disk space for one version no matter hopw many copies you have, this is done using a file link, links are common in the system directories, and are used for configuration as werll as dynamic libraries.

Hard Link
Assume that the current working directory is /home/jane. We wish to create a link 'hardfile2' within the subdirectory projects from the file 'hardfile'.
% date > hardfile (create the file)
% ls -l -rwx-xr-x. 1 jane users 605 Nov 18 12:25 hardfile
% ln hardfile projects/hardfile2
% ls -l projects/hardfile2 -rwx-xr-x. 2 jane users 605 Nov 18 12:25 projects/hardfile2
The file hardfile and its link hardfile2 are indistinguishable, if hardfile is updated then hardfile2 is updated, notice the link number has increased to 2- this owuld occur in both listings.
Soft Links(symbolic links)
Let us assume that the current working directory is /home/john and we wish to create a link 'softfile2' within the subdirectory projects to the file softfile. Notice the -s flag
% date > softfile (create the file)
% ls -l -rwx-xr-x. 1 john users 605 Nov 18 12:25 softfile
% ln -s /home/john/softfile projects/softfile2
% ls -l projects/softfile2 lrwx-xr-x. 1 john users 605 Nov 18 12:25 projects/softfile2 -> /home/john/softfile
Nitce the appended pathname on the long listing, the link number has not chantged but the permissions show an l at hte beginning of the long listing ratehr than a '-', again any ujpdates in softfile will be reflected in softfile2



A hard link points to the same disk block(s)[inode] as the original file. Still works when orginal file is deleted.
A soft link points to the original file, like a windows shortcut. Left hjanging if orgional file is deleted.

PROCESSES
Processes are running programs. They have there own ID(pid, program id i assume)
Some processes are part of the filesystem and can be found. Some processes are special aand cannot be found, and these are usually described [brackets]. The INIT process is the boss process in linux.

> ps aux
USER PID %CPU %MEM VSZ RSS TTY STAT START TIME COMMAND
root 1 0.1 0.8 1480 496 ? S 12:57 0:00 init [5]
root 2 0.0 0.0 0 0 ? SWN 12:57 0:00 [ksoftirqd/0]
root 3 0.0 0.0 0 0 ? SW< 12:57 0:00 [events/0]
root 4 0.0 0.0 0 0 ? SW< 12:57 0:00 [khelper]
root 16 0.0 0.0 0 0 ? SW 12:57 0:00 [kjournald
root 527 0.0 0.9 1464 576 ? S 12:58 0:00 syslogd -m 0
rpc 553 0.0 0.9 1544 584 ? S 12:58 0:00 portmap
rpcuser 573 0.1 1.3 1644 812 ? S 12:58 0:00
rpc.statd root 658 0.4 2.4 3656 1484 ? S 12:58 0:00 /usr/sbin/sshd
gordon 15521 0.0 0.1 3992 760 pts/1 R 20:41 0:00 ps aux

State codes
Standard Codes
-D uninterruptible sleep(usually I/O)
-R runnable(on run queue)
-S sleeping
-T traced or stopped
-W paging
-X dead
-Z a defunct(“zombie”) process
Additoinal codes
- W has no resident pages
- < high priority process

- N low priority task(nice)
- L has pages locked into memory(for real time and custom IO)

Processes relationships

Processes form “trees” of parentage. All processes have the parent INIT. If a process starts another process, that new process has a parent of the old process. I now run pstree, in the bash shell, after logging into the machine using ssh(controlled by sshd)

> pstree

```
[root@host-1-97 ~]# pstree
systemd─┬─ModemManager───2*[{ModemManager}]
        ├─NetworkManager─┬─dhclient
        │                └─3*[{NetworkManager}]
        ├─2*[abrt-watch-log]
        ├─abrtd
        ├─accounts-daemon───2*[{accounts-daemon}]
        ├─acpid
        ├─alsactl
        ├─at-spi-bus-laun─┬─dbus-daemon───{dbus-daemon}
        │                 └─3*[{at-spi-bus-laun}]
        ....
        ├─smartd
        ├─sshd───sshd───bash───pstree
        ├─systemd-journal
        ├─systemd-localed
        ├─systemd-logind
        ├─systemd-udevd
        ├─tuned───4*[{tuned}]
        └─upowerd───2*[{upowerd}]
```

/proc

Processes are represented as files in /proc

They appear as directory with a name equal to the PID of the process. They have things in the directory which define the process in question.

> ls –l /proc/668

```
-r--------.  1 root root 0 Sep 21 16:32 auxv
-r--r--r--.  1 root root 0 Sep 21 16:31 cmdline
lrwxrwxrwx.  1 root root 0 Sep 21 16:32 cwd -> /
-r--------.  1 root root 0 Sep 21 16:32 environ
lrwxrwxrwx.  1 root root 0 Sep 21 16:32 exe -> /usr/sbin/sshd
dr-x------.  2 root root 0 Sep 21 16:32 fd
-r--r--r--.  1 root root 0 Sep 21 16:32 maps
-rw-------.  1 root root 0 Sep 21 16:32 mem
-r--r--r--.  1 root root 0 Sep 21 16:32 mounts
lrwxrwxrwx.  1 root root 0 Sep 21 16:32 root -> /
-r--r--r--.  1 root root 0 Sep 21 16:31 stat
-r--r--r--.  1 root root 0 Sep 21 16:32 statm
-r--r--r--.  1 root root 0 Sep 21 16:31 status
dr-xr-xr-x.  3 root root 0 Sep 21 16:32 task
-r--r--r--.  1 root root 0 Sep 21 16:32 wchan
```

> ls –l /proc/668/fd

```
lrwx------.  1 root root 64 Sep 21 16:32 0 -> /dev/null
lrwx------.  1 root root 64 Sep 21 16:32 1 -> /dev/null
lrwx------.  1 root root 64 Sep 21 16:32 2 -> /dev/null
lrwx------.  1 root root 64 Sep 21 16:32 3 -> socket:[4230]
```

Files which that process has open
0,1,2 are STDIN,STDOUT,STDERR
668:SSHD listening on a socket for people logging in with ssh
> sleep 20 > /tmp/hia & [1] 854
> ls –l /proc/854
> ls –l /proc/854/fd
Daemons
A daemon is a process started when you boot which runs in the background. Not all things started when booting stay running(they set something up and then die) to help us, daemons usually have a name which ends with a “d”
> top

```
top - 16:03:17 up  1:04,  1 user,  load average: 0.00, 0.00, 0.00
Tasks:  35 total,   2 running,  33 sleeping,   0 stopped,   0 zombie
Cpu(s):  0.0% us,  0.0% sy,  0.0% ni, 100.0% id,  0.0% wa,  0.0% hi,  0.0% si
Mem:     59764k total,    52308k used,     7456k free,     6192k buffers
Swap:   205816k total,        0k used,   205816k free,    32472k cached

  PID USER      PR  NI  VIRT  RES  SHR S %CPU %MEM    TIME+  COMMAND
  807 root      16   0  1624  728 1412 R  0.0  1.2   0:00.02 in.telnetd
  934 root      16   0  1828  872 1628 R  0.0  1.5   0:00.00 top
```

SYSLOG
The syslogd daemon is your friend. It helsp otehr daemons record what is going on into a file. On the website you can click on syslog output and see what syslogd has noticed. This output known as the syslog, can also be seen from the prompt using “dmesg”

>who
root pts/0 Sep 21 15:59 (hub1-gw)
I am root, logged on from hub1-gw. My session is linked to a device which handles my screen and keyboard, called pts/0, this refers to /dev/pts/0

> ls –l /dev/pts/0
crw--w----. 1 root tty 136, 0 Sep 21 16:49 /dev/pts/0
This is a character device. The person connected via it alwaysb owns it. There are no sizes with block or char devices. 136 is the major devvice nmumber, 0 is the minor device number

>man ps
Processes status

Networked Services Basic Administration Concepts part 1
Disk in linux - disk storage is a layered system
Disk -> Partitions ->logical volume manager -> filesystem
->filesystem
Disks
In linux a disk is often a SATA disk, and is identified by:
– /dev/sda – lowest numbered SCSI device
– /dev/sdb – next lowest SCSI device

The CDROM drive, which is also a block storage device, frequently has a name /dev/sr0
– For convenience /dev/cdrom is a softlink to the actual cdrom device $ ls –l /dev/cdrom
lrwxrwxrwx. 1 root root 3 Sep 25 13:10 /dev/cdrom -> sr0

Partitions
Rather than use the whole disk for one purpose. Split disk up into chunks. The chunks are known as partitions. Partitions can be primary or secondary. This is partially a hang-over from when DOS could only handle 4 partitions.

```
>  sfdisk -l /dev/sda

Disk /dev/sda: 19449 cylinders, 255 heads, 63 sectors/track
Units = cylinders of 8225280 bytes, blocks of 1024 bytes, counting from 0

   Device Boot Start     End   #cyls    #blocks   Id  System
/dev/sda1   *      0+   1274-   1275-  10240000   83  Linux
/dev/sda2       1274+   3824-   2550-  20480000   82  Linux swap / Solaris
/dev/sda3       3824+  19449-  15625- 125506560   83  Linux
/dev/sda4          0       -       0          0    0  Empty
```

Common Partitions
At a minimum a linux box has - “/”, the partitions which holds the main directory tree, such as the operation system files.
“/boot” a partition which stays close toe the start of the disk structure, designed to be easily accessible from a variety of BIOS configurations and disk formats, used for bootstraping.
“Swap” an area for holding virtual memory which has been swapped out of main memory.
Options include
“/home” to keep users and systems files physcially seperate. “/var” as some feel that /var is changed at a higher rate than other partions thus there may be more security in formatting is seperately.

Logical Volume Manager LVM
If you want to change your partitions after installing, then you are doing something very risky. It is possibnle to destroy every partion by damaging the partition table. Often the current partitions use 100% of the disk, so repartitioning to make something bigger means shrinking other partitions. Formatted partitions dont like being shrunk and need to be compressed and migrated before being shrunk. LVM aand other tools tries to make this simpler, by adding dynamic run-time controls to partition management
LVM
$ lvdisplay
--- Logical volume ---
LV Path /dev/centos_lvm/root
LV Name root
VG Name centos_lvm
LV UUID xnNIQ2-ct1m-UqaR-BkjO-GrNI-FUwt-e2ciYM
LV Size 6.51 GiB

LVM Management
You can do things dynamically but you still have to care.
Create a new parition in an LVM? $ lvcreate –L 2G –n newstuff centos_lvm
Shrink an existing partition? $ lvreduce –L-1G –n mypartition centos_lvm

Add a new harddrive effectrively glue its space onto lvm andf make the dynamic area bigger
$ vgextend centos_lvm /dec/sdb1
To make resizing work, you still have to carefully resize the partition format, but at least there is no danger in deleteing delting all the partions accidently
Parition format
When you format a parititon for sue, in say EXT3, or NTFS, of XFS, the partition gets a block device id. This uniquely(hopefully) identifies the partition. In the past you had to identify partitions by their number and disk device name, such as partition 2 on sda being /dev/sda2. Better to identify them by block id, so if partitions are mvoed(say by lvm) or disks rearranged(so sda becomes sdb), you can still identify a partition easily, reliability, and consistently…
Block identifiers
$ blkid

```
/dev/sda1: UUID="f3b744e4-e754-4842-93d1-43b06de64b66" TYPE="xfs"
/dev/sda2: UUID="kIogiH-f548-AELA-NMVK-sbr7-9u7j-2K6MDz" TYPE="LVM2_member"
/dev/sdb1: UUID="971d09b6-8ce8-49c7-9ec9-16b0155f42cf" TYPE="swap"
/dev/mapper/centos_lvm-root: UUID="b66fdf9b-16f0-4648-9663-536881db0ab1" TYPE="xfs"
```

> cat /etc/fstab
When the system boots the fstab file tils the kernal what filesystems to load

```
/dev/mapper/centos_lvm-root                /      xfs     defaults        1 1
UUID=f3b744e4-e754-4842-93d1-43b06de64b66 /boot  xfs     defaults        1 2
UUID=971d09b6-8ce8-49c7-9ec9-16b0155f42cf swap   swap    defaults        0 0
```

> df

```
Filesystem                  1K-blocks    Used Available Use% Mounted on
/dev/mapper/centos_lvm-root   6813696 4100528   2713168  61% /
devtmpfs                       241940       0    241940   0% /dev
tmpfs                          250952      80    250872   1% /dev/shm
tmpfs                          250952    4752    246200   2% /run
tmpfs                          250952       0    250952   0% /sys/fs/cgroup
/dev/sda1                      508588  136588    372000  27% /boot
```

“Df -h” is also useful , translating bytes in MB or GB as appropriate

Disk usage
If you want to find out how muc hdisk space a directory is using the du command does this easily

```
$ du -s /usr/lib
477464  /usr/lib
$ du -sh /usr/lib
467M    /usr/lib
```

-s is useful, otherwise it tills you abuot all subdirecotries
-h puits it into human readable form.

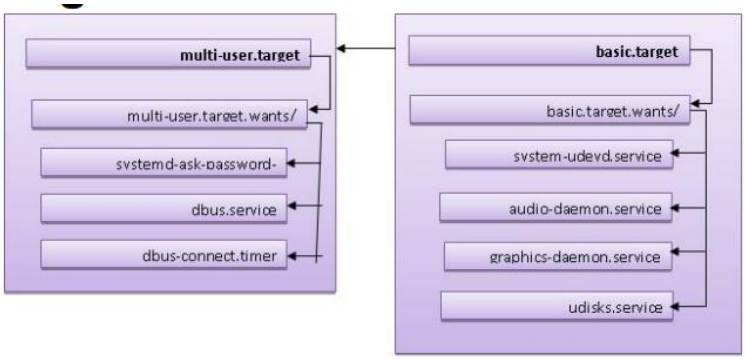
Linux Boot Process
Booting to kernel

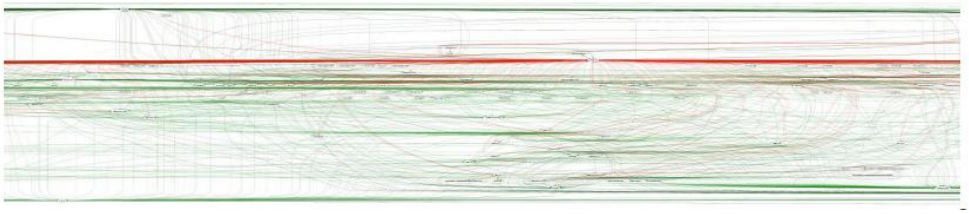
From switch on: – PC BIOS selects a boot disk – BIOS loads the boot block and executes it. – This loads a stage 1 boot loader. – Stage 1 loads stage 2 loader. – Linux loader (e.g. Grub, lilo) runs – Operator selects from loader menu – Kernel loaded with device ramdisk

Startup Commands

As linux boots, it runs various systems scripts. In redhat/centos/fedora, this is controlled largely by systemd. - in the olden days this was controlled by init.d scripts, based on system v, systemd forms a tree of dependencies: start the network before starting sshd, start the firewall after the network, the internals of systemd are very comlpex, so best to alwayus use teh system commands control how systemd runs.

Target tree





Systemd.analyze

Controlling services in systemd

There are many services you may wish to control, such as sshd, apache, databases, etc. Things which are services are named “something.service” such as sshd.service. Control is via systemctl. Start a service: systemctl start sshd.service. Stop a service: systemctl stop sshd.service. Restart sshd: syustemctl restart sshd.service, reload sshd:systemctl relaod sshd.service.

Make things start when you boot

Start a service: systemctl start sshd.service - the sshd service will run only till you reboot. If you want something to run every time you rebot then enable it - systemctl enable sshd.service

Change your mind and want it disabled next time you boot? -systemctl disable sshd.service

Stop a service: systemctl stop sshd.service.

Status

The informatoin about a service is stored in a few locations.

You can do that yourself, but systemctl can build the info for you, systemctl status sshd.service

```
[root@host-19-17 grub.d]# systemctl status sshd.service
sshd.service - OpenSSH server daemon
   Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled)
   Active: active (running) since Mon 2014-09-29 14:55:19 BST; 1h 23min ago
  Process: 1014 ExecStartPre=/usr/sbin/sshd-keygen (code=exited, status=0/SUCCES
S)
 Main PID: 1023 (sshd)
   CGroup: /system.slice/sshd.service
           └─1023 /usr/sbin/sshd -D

Sep 29 14:55:19 host-0-0.linuxzoo.net systemd[1]: Started OpenSSH server daemon.
Sep 29 14:55:20 host-0-0.linuxzoo.net sshd[1023]: Server listening on 0.0.0.0...
Sep 29 14:55:20 host-0-0.linuxzoo.net sshd[1023]: Server listening on :: port...
Sep 29 14:55:36 host-19-17.linuxzoo.net sshd[1306]: Accepted password for roo...
Sep 29 14:57:30 host-19-17.linuxzoo.net sshd[1362]: Accepted password for roo...
Sep 29 15:13:20 host-19-17.linuxzoo.net sshd[1665]: Accepted password for roo...
Sep 29 15:17:37 host-19-17.linuxzoo.net sshd[1746]: Accepted password for roo...
Sep 29 16:02:23 host-19-17.linuxzoo.net sshd[2409]: Accepted password for roo...
Hint: Some lines were ellipsized, use -l to show in full.
[root@host-19-17 grub.d]#
```

Configuration file: /usr/lib/systemd/system/sshd.service

```
[Unit]
Description=OpenSSH server daemon
After=syslog.target network.target auditd.service

[Service]
EnvironmentFile=/etc/sysconfig/sshd
ExecStartPre=/usr/sbin/sshd-keygen
ExecStart=/usr/sbin/sshd -D $OPTIONS
ExecReload=/bin/kill -HUP $MAINPID
KillMode=process
Restart=on-failure
RestartSec=42s

[Install]
WantedBy=multi-user.target
```

System V init.d

This is still in use in some parts of centOS and in linux in general. Orgionally there were “runlevels” each numbered and associated with a concept like “boot with graphical interface”, “boot with no networking”, etc. booting with a GUI is considered runlevel 5. By default in sysv, runlevel 5 is considered normal. What runs at runlevel 5 is controlled by soft links in /etc/rc5.d - each link indicates what runs or not at this runlevel. During boot or shutdown the system transition between runlevels. During boot it enters runlevel 5 5, and during shutdown it enter runlevel 6. It softlinks to scripts in /etc/init.d, which control each service. The name of the link indicates what happens in that runlevels

> ls –l rc5.d
lrwxrwxrwx. 1 root root 20 Sep 8 09:20 K50netconsole -> ../init.d/netconsole
lrwxrwxrwx. 1 root root 17 Sep 8 13:06 S10network -> ../init.d/network
lrwxrwxrwx. 1 root root 17 Sep 8 09:35 S20iprinit -> ../init.d/iprinit
lrwxrwxrwx. 1 root root 19 Sep 8 09:35 S20iprupdate -> ../init.d/iprupdate
lrwxrwxrwx. 1 root root 17 Sep 8 13:06 S21iprdump -> ../init.d/iprdump

S/K priority service-name
S10network
Ls -l s10network
lrwxrwxrwx. 1 root root 17 Sep 8 13:06 S10network -> ../init.d/network
Starts at priority 10 - runs after 9 and before 11
S indicates the system shoudl start networking at this runlevel
K indicates the system should stop something isntead
So rc6.d holds things like: lrwxrwxrwx. 1 root root 17 Sep 8 13:06 K90network -> ../init.d/network
The syslog
Dmesg shows the messages which have recently been generated by the kernel…
This and service messages are collated by the syslogd.
It sorts messages out according to a config file in /etc, and then categorises the messages and stores them somewhere in /var/log/
For isntance --systemd messages end up in “messages” , sshd messages end up in “secure”, web server diagnostics will end up in /var/log/httpd/
The location of some files and their function does change between distributions, you can always use “systemctl status to locate recent messages.
The xinetd super-daemon

Some of the services'(sshd) are processes, they start running from an rc script. They wait on their own for comms, they terminate only when the machine does down. Some people say this wastes resources, the super server concept was born. The super server concept was born.
XINETD
Einetd waits for requests from the internet. From the requests it works out whjat program would like to with that request. It then starts that program running and gives it the waiting requests. In this wya resources are only used if someone actually requests access to a particular service.

Systemd supports this type of service too, but without XINETD - however rather than "service" this is known as socket
For out virtual machines, telnet has been setup to use this - telnet.socket, when connections arrivate it triggers in turn telnet.service for that connection.

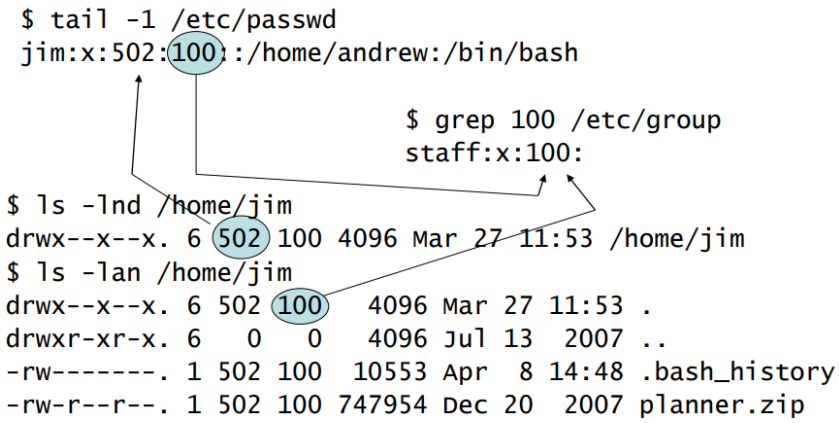
Terminating a process
If you know the process id(the pid) of a program you can termite it quickly and easily, you send it a message(a signal) to tell it to end. The message to end now is called SIGKILL.
> ps aux | grep sshd
root 1796 …… /usr/sbin/sshd
> kill –s SIGKILL 1796

Networked Services Basic Administration Concepts Part 2
User Management
Manual creation
User entries in passwd, shadow, group, gshadow. Home directory in /home. Copy basic .files into their home directory. Make new user own their own ddirectory and files.
> adduser gordon
This does all the magic for you. It copies the default .files from /etc/skel/
> ls –a /etc/skel/
.bash_logout .bash_profile .bashrc .gtkrc .kde
In bash, .bashrc is executed in non-login shell, and .bash_profile in a login shell.#
Skel files
These files are the basic .Files created for a new user. Users are free to edit these when they log in. this allows them to control their own path, env and other settings(such as aliases)
However, if you install a new package which needs something set for each user at login, editing all these copies by hand would be tiresome.
> ls /etc/profile.d
colorls.csh gnome-ssh-askpass.csh krb5.csh less.csh vim.csh colorls.sh gnome-ssh-askpass.sh krb5.sh less.sh vim.sh glib2.csh kde.csh lang.csh qt.csh which-2.sh glib2.sh kde.sh lang.sh qt.sh
If you log in with bash, all the .sh files are executed before your .files
If you log in with csh, all the .csh files are executed before your .files
> cat /etc/profile.d/vim.sh
if [-n "$BASH_VERSION" -o -n "$KSH_VERSION" -o -n "$ZSH_VERSION"]; then # for bash, pdksh and zsh, only if no alias is already set alias vi >/dev/null 2>&1 || alias vi=vim fi

if this is bash, and you have not set an alias for “vi”, then set one to run “vim” when you type “vi”.

Example
• Create a user jim, in group staff • But how to set the group? • You could do: $ man adduser • Usually commands also take the flag “-h” $ adduser –h adduser: invalid option -- h Usage: useradd [options] LOGIN Options: -b, --base-dir BASE_DIR base directory for the new user account … -g, --gid GROUP force use GROUP for the new user account..
$ adduser jim –g staff $ tail -1 /etc/passwd jim:x:502:100::/home/jim:/bin/bash $ grep 100 /etc/group staff:x:100:

Moving a uid or gid



Useful Commands
$ chown jim.staff filename
$ chown jim filename
$ chgrp staff filename
When a User logs in
When a user logs in, the appropriate . files arre executed (.login, .cshrc,) if you want to change to a different user, you could log out and log in again, or you could do
Su - gordon(change to the gordon user)
Su - (change to root)
Without the “-” you still cahnge users, but the . scripts dont get executed
To go back to the previous user, press ctrl-d
*

End System Networking
Linux Networking
Linux is a capable networking platform, runs many server applications, so is often seen as a prime platform for server applications, it has extensive level 2 and 3 networking support, it supports multiple network connections.
In this course the old network commands, such as ifconfig and route, are deprecated. They may be removed from linux distributions at any time.

Default Networking

Linux is a system which needs networking in order to work correctly, even a system with no network has networking, the basic network is the loopback network, every computer has an ip on the loopback network named localhost.
> telnet localhost > telnet 127.0.0.1 > ping localhost

Localhost
Ip of localhojst is 127.0.0.1
Operates as a true network, and anything which can be done on a network in linux can operate on the localhost network. Linux operates a priority networking system, and localhost has the highest priority. If a packet can be delivered using localhost then it will always be delivered with localhost

The localhost network device
“Lo” is often thought of as the localhost network device however it is rarely implemented as a /dev device. However as the commands which expect a network device will take lo as a device name. It is handled internally in the kernel.
$ip link show lo
1: lo: mtu 65536 qdisc noqueue state UNKNOWN mode DEFAULT link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
LOOPBACK its a loopback device. UP - the network is running at layer 2. LOWER_UP - Active at layer 1.
Qdisk - Queueing discipline - how queued things are processed.

The Network Device
In many systems /dev/eth* is the ethernet network device. In such systems with only one network connections, /dev/eth0 is the standard device name. Some distributions are sometimes renaming eth0..n to reflect the hardware bus number of the device. This makes the name the same no matter how many hardware devices are plugged in later, on my server i have enp3s0 and enp2s0.
A basic network needs - ip number of the host, netmask for the network, gateway ip for the gateway, broadcast address.
The modern way to specify an IPv4 is the normal IP number and a /n value informing you of the netmask. 10..0.1.20/24. This indicates a ip of 10.0.1.20, a netmask of the first 24 bits(255.255.255.0), senibly a broadcast of 10.0.1.255, sensibly a gateway of 10.0.1.254

$ip link show ens3
2: ens3: mtu 1500 qdisc pfifo_fast state UP mode DEFAULT qlen 1000 link/ether 00:0c:11:00:13:90 brd ff:ff:ff:ff:ff:ff
At layer 2, it supported broadcast and multicast, queing deffaults to a prioritised fifo queuing discipline. The mac address is 00:0c:11:00:13:90
dynamic ens3 valid_lft 863827sec preferred_lft 863827sec inet6 fe80::20c:11ff:fe00:1390/64 scope link valid_lft forever preferred_lft forever
Now caan see the layer 3 information.. ip , broadcast. Dynamic indicates the address is learned via dhcp.

$ ip –s link show ens3
2…
RX: bytes packets errors dropped overrun mcast

44073	425	0	0	0	0
TX: bytes	packets	errors	dropped	carrier	collsns
71632	336	0	0	0	0

Various statistics can be obtained which may help with diaagnostics

Ethernet Errors
Difficult to find out exact meanings, but it is likely that:
Errors - CRC Error in packet
Dropped - Kernel buffers overflowed
Overruns – Card buffer overflowed
Frame – Frame length not a multiple of 8 bits
Carrier – Probably a fault in the card
Collisions – tx collided with another frame

ip address add IFADDR dev DEVICE
The ifaddr can ccontain information for setting many things, includeing:ip, broadcast address, netmask
Ip address add 10.0.50.10/24 broadcast 10.0.50.255 dev eth0

Interface selection
The ip command describes how each device should be used. The selection of the interface, and the rules as to what interface is used for each packet, are stored in the routing table. Linux has an advanced rule based routing table, we will only be doing simple routing, we will use thje “ip route” command for routing table manipulation. It is possible to have a large number of routing tables - which table to use depends on rules, by default, the rules are simple.

> ip rule shows
0: from all lookup local 32766: from all lookup main 32767: from all lookup default
Rules point to tables, which are like subroutines in a program. The number is the priority, this case table local is first, then main, then default. If the network packet is handled in a particular table, it is not passed on to any other tables.
> ip route show table local
broadcast 10.0.19.16 dev ens3 proto kernel scope link src 10.0.19.17
local 10.0.19.17 dev ens3 proto kernel scope host src 10.0.19.17
broadcast 10.0.19.23 dev ens3 proto kernel scope link src 10.0.19.17
broadcast 127.0.0.0 dev lo proto kernel scope link src 127.0.0.1
local 127.0.0.0/8 dev lo proto kernel scope host src 127.0.0.1
local 127.0.0.1 dev lo proto kernel scope host src 127.0.0.1
broadcast 127.255.255.255 dev lo proto kernel scope link src 127.0.0.1

So all packets are tested in the local table first. If a packet is obviously handledf using what we know about a link, then make the obviousl choice.

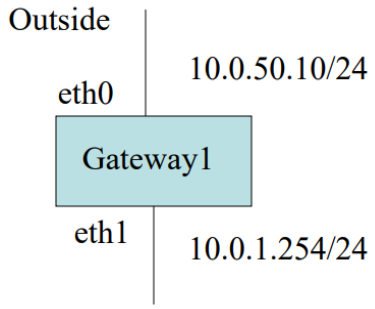
> ip route show table main
default via 10.0.19.22 dev ens3 proto static metric 1024
10.0.19.16/29 dev ens3 proto kernel scope link src 10.0.19.17

For an end network PC, packets are either for the local network or must be forwarded to the gateway. Rules move from most specific to least specific.
1. Destination in 10.0.19.16/29, send directly via ens3.
2. Other destination? - ask 10.0.19.22 to forward the packet.

Route
For a simple example of: 10.0.50.10/24
$ ip addr add 10.0.50.10/24 broadcast 10.0.50.255 dev eth0
$ ip route append 10.0.50.0/24 dev eth0 table main
$ ip route append default via 10.0.50.254
Table main is the default, so can be left out of ip route. By default, the default route works out which device to send on.

Linux as a “Router”
If linux has more than 1 network connection, it can perform layer 3 routing, just like a cisco router. Cisco router often have only 2 or 3 network connections, and it is easy to build a PC to replicate this, cisco argue that their routers are far superior. Physically cisco routers have ASICs that make routing faast, with various hardfware “things” to cache decisions and process data. For generic routing cisco will most likely outperform linux. But physically routers may need other things, like services, firewalls, IDS and IPS, and advanced traffic management features.such features may be better handled by linux, but with a performance/cost ratio lower than hardware devices.

Configuration
Multiple networks are no different from a single network configurations. You need “ip address” for each interfaces. You need a route for each interface, some configuration approaches may add route information authomatically. But you should still be able to do it manually, and understand what is going on, as the automation may not be right. It is also good for exam questions!, in the tutorials the routes do not seem to be automatically added. You also need 1 default route.
GGeneral Rule: if you have “n” ethernet defvices, you need “n” ip route commands(one per device) plus 1 route for the default gateway.



Add this example
> ip addr add 10.0.50.10/24 broadcast 10.0.50.255 dev eth0
> ip route append 10.0.50.0/24 dev eth0
> ip addr add 10.0.1.254/24 broadcast 10.0.1.255 dev eth1
> ip route append 10.0.1.0/24 dev eth1
> ip route append default via 10.0.50.254

> ip route show
10.0.50.0/24 dev eth0 scope link
10.0.1.0/24 dev eth1 scope link
default via 10.0.50.254 dev eth0

The netmask
The netmask can be any size from /0 to /32. Perhaps you considered only /8,/16,/24 masks. These are fixed-length masks, matching the IP type(like class A, B,etc). Complex networks use variable-length subnet masks.

VLSM
Variable length subnet masks: subdivide the host part of the network mask into smaller pieces. Each subdivision has its own network. So if you need to run 2 networks, but only have 10.1.1.0/24, you can create 2 networks as: – 10.1.1.0/25 – 10.1.1.128/25. Remember that first and last host is reserved for “network” and “broadcast”. Thus you cannot use 10.1.1.0 or 10.1.1.127 or 10.1.1.128 for host addresses.

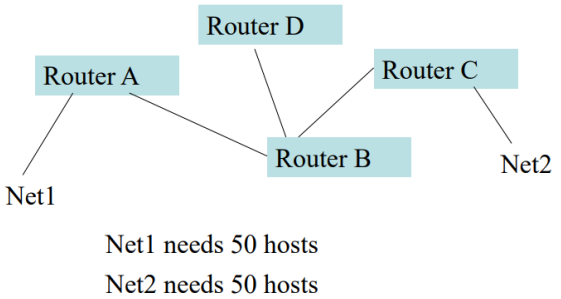
VLSM is “borrowing bits”
Problem: You need 5 networks, but you only have 10.10.10.0/24. • You cannot split into an number of networks which is not a power of 2 (2,4,8,16,etc), so split into 8. • 8 needs 3 bits in binary (000-111 is 8 combinations) • So borrow 3 bits from /24, making it /27. • The new network numbers are: – 10.10.10.0/27 10.10.10.32/27 – 10.10.10.64/27 10.10.10.96/27 – 10.10.10.128/27 10.10.10.160/27 – 10.10.10.192/27 10.10.10.224/27

VLSM for minimum hosts
Sometimes you have a problem which states that you need n hosts per network. • Consider the example of 10.1.1.0/24, where you need to divide your network into as many subnets as possible, where each subnet can hold at least 10 hosts. • Increase “10” by 2, then increase to the next power of 2 (i.e. 16). • 16 needs 4 bits (0000-1111 is 16 combinations). • Take 32-4, giving 28. Network is 10.1.1.0/28, or: – 10.1.1.0/28, 10.1.1.16/28, 10.1.1.32/28, etc.

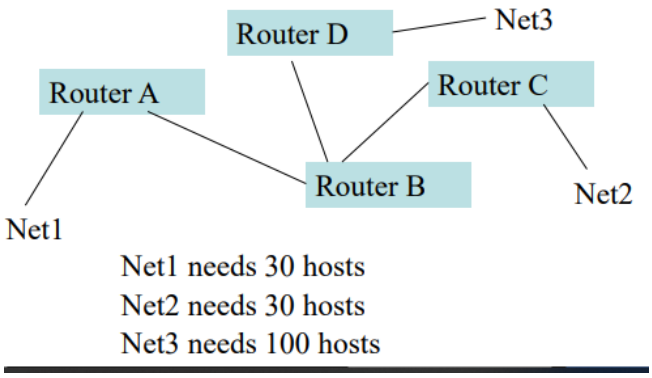
Broken VLSM
Some legacy systems dont understand VLSM(eg RIP), sometimes called the “subnet zero” problem. This leads to 2 points of confusion concerning the first and last network: with 10.10.10.0/24 split into /27, networks 10.10.10.0/27 and 10.10.10.224/27 cause problems. For 10.10.10.0/27, 10.10.10.0 is the network number, and 10.10.10.255 is the broadcast address. But in VLSM, it’s the network number for network 1, and the broadcast for network 8.

P2P netmask
A point to point network is a little weird. – 10.0.0.14/32 – Netmask 255.255.255.255 – Broadcast 10.0.0.255 – Gateway is likely to still be 10.0.0.254. The gateway IP can be reused multiple times on each p2p link without difficulties.
• Really small netmasks > 1 IP … • 10.0.0.5/30 – 2 bits unset thus only 4 IPs in this net – IPs are 10.0.0.4,10.0.0.5,10.0.0.6,10.0.0.7 – Broadcast will be highest ip, 10.0.0.7 – The network has its own address (all bits zero) which reserves 10.0.0.4 for the network. – Max-1 is often

the gateway, 10.0.0.6 – Only 1 IP for host, 10.0.0.5 • Other than p2p, biggest netmask must be /30.



• 50 hosts suggests 6 bits, leaving 2 bits, /26 • This provides 4 networks. • However there are 5 networks: – Net1 – Net2 – Router A-B – Router B-C – Router B-D
Solution is to divide up one /26, and use that for routerrouter links. • For Net1+2 50 hosts suggests 6 bits, leaving 2 bits, /26 • For routers, 2 hosts suggests 2 bits or /30. • Use 10.1.1.0/26 for Net1, 10.1.1.64/26 for net2. • Split 10.1.1.128/26 into multiple /30 links: – Net1 – 10.1.1.0/26 – Net2 – 10.1.1.64/26 – Router A-B – 10.1.1.128/30 – Router B-C – 10.1.1.132/30 – Router B-D – 10.1.1.136/30



Solution: • 10.1.1.0/24 gets split into: – Net3 – 10.1.1.0/25 – Net1 – 10.1.1.128/27 – Net2 – 10.1.1.160/27 – Router A-B – 10.1.1.192/30 – Router B-C – 10.1.1.196/30 – Router B-D – 10.1.1.200/30

Linux Switch
A linux box can also operate as a layer ⅔ device. Here multiple ethernet cards are configured as layer 2 devices(mac address but no ip). They are then bridged together to form an intelligent switch. Hardware switches have custom logic to perform switching, and linux boxes do this all in software.

Example bridge configuration # ********** Create a bridge interface and it is called br1 brctl addbr br1 # ********** Add physical interfaces to the bridge interface brctl addif br1 eth0 brctl addif br1 eth1 # ********** Reset IP interface ifconfig eth0 0.0.0.0 ifconfig eth1 0.0.0.0 #Bring up the bridge ifconfig br1 up # ********** Set IP address of the bridge ifconfig br1

192.168.1.10 netmask 255.255.255.0 up # ********** Set IP default gateway route add default gw 192.168.10.1

Network troubleshooting
Layered Approach
Check layer 1 - is it wired up
Check layer 2 - Ethernet framing ok?, layer 2 addressing?
Check layer 3 - ip addresses and routes, ping, traceroute
Check layer 4 - nmap

Linux ARP cache
With an ethernet device the kernel must perform an ARP lookup. ARP is expensive, so the result is cached. /proc/net/arp is the arp cache record.

```
> cat /proc/net/arp
IP address … HW address .. Device
146.176.166.254 00:08:7c:6e:90:00 eth0
146.176.166.2 00:e0:81:26:31:06 eth0
> ping 146.176.166.6
> cat /proc/net/arp
IP address … HW address .. Device
146.176.166.254 00:08:7c:6e:90:00 eth0
146.176.166.2 00:e0:81:26:31:06 eth0
146.176.166.6 00:e0:81:25:c7:35 eth0
```

nmap

```
$ nmap linuxzoo.net
PORT STATE SERVICE
22/tcp open ssh
23/tcp open telnet
53/tcp open domain
80/tcp open http
81/tcp open host2-ns
123/tcp closed ntp
5900/tcp closed vnc
5901/tcp closed vnc-1
5902/tcp closed vnc-2
5903/tcp closed vnc-3
```

Netstat - another mointoring tool, lots of options

```
$ netstat -al | grep LISTEN | grep tcp
tcp 0 0 *:http *:* LISTEN
tcp 0 0 *:ssh *:* LISTEN
tcp 0 0 *:https *:* LISTEN
$ netstat -n | head -4
Active Internet connections (w/o servers)
 Proto Recv-Q Send-Q Local Address Foreign Address State
tcp 1 0 127.0.0.1:64359 127.0.0.1:631 CLOSE_WAIT
```

tcp 0 0 146.176.162.6:22 146.176.16:59160 ESTABLISHED
Not sure about port ":22"?
$ grep '22/tcp' /etc/services
ssh 22/tcp # SSH Remote Login Protocol
bpjava-msvc 13722/tcp # BP Java MSVC Protocol

Subnet Cheat Sheet

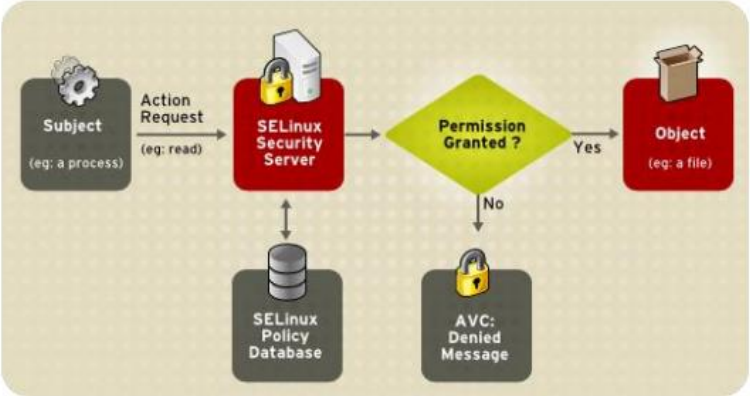
CIDR	Total # of Addresses	Hosts	Netmask
/30	4	2	255.255.255.252
/29	8	6	255.255.255.248
/28	16	14	255.255.255.240
/27	32	30	255.255.255.224
/26	64	62	255.255.255.192
/25	128	126	255.255.255.128
/24	256	254	255.255.255.0
/23	512	510	255.255.254.0
/22	1,024	1,022	255.255.252.0
/21	2,048	2,046	255.255.248.0
/20	4,096	4,094	255.255.240.0
/19	8,192	8,190	255.255.224.0
/18	16,384	16,382	255.255.192.0
/17	32,768	32,766	255.255.128.0
/16	65,536	65,534	255.255.0.0
/15	131,072	131,070	255.254.0.0
/14	262,144	262,142	255.252.0.0
/13	524,288	524,286	255.248.0.0
/12	1,048,576	1,048,574	255.240.0.0
/11	2,097,152	2,097,150	255.224.0.0
/10	4,194,304	4,194,302	255.192.0.0
/9	8,388,608	8,388,606	255.128.0.0
/8	16,777,216	16,777,214	255.0.0.0
/7	33,554,432	33,554,430	254.0.0.0
/6	67,108,864	67,108,862	252.0.0.0
/5	134,217,728	134,217,726	248.0.0.0
/4	268,435,456	268,435,454	240.0.0.0
/3	536,870,912	536,870,910	224.0.0.0
/2	1,073,741,824	1,073,741,822	192.0.0.0
/1	2,147,483,648	2,147,483,646	128.0.0.0
/0	4,294,967,296	4,294,967,294	0.0.0.0

Netmask cheat sheet stuff
Also remmeber that routing table goes from most specific to least specific(as in the higher the / , if you have a /29 it will be used rather than a /24 even if it applies to that packet) not the order that it is done in

SELinux
Linux security model

Linux by default uses DAC for its file access model, DAC Discretionary Access Controls. DAC is based on UID or GID and uses file access controls based on RWX - DAC also includes finer-grained ACLs.
DAC has some issues - if permission is granted to a user then this is independent of user behaviour. So for instance if a good process was hacked in some way, it can continue to access what it could before even if it was operating in a different way.
ACLs are time consuming to maintain, Grant and revoke of permissions requires finding all the files related to the change, cannot easily stay negative statements(ie what you are not allowed to access). Non file based permissions are tricky to manage, eg what ports can be opened by a process, what permissions do child processes have, or what processes can a process talk to.

SELinux
Orgionally developed by the NSA(national security Agency), is an implemention of the Flask operating system security architecture.
Implements MAC - Mandatory Access Control, MAC is a centrally managed security policy based on objects
The idea is to give processes “least privilege” ie only exactly what the process needs to get the job done. Each file is given an SELinux Label using xattr. Each process is labelled. Selinux holds rules abuot what process labels can do to each file label



Modes of Operation
SELinux can be run in - Enforcing Mode(default), Permissive Mode, Disabled
SELinux can also operate in a few types of enforcement, Strict, Targeted(default), minimum,mls.
Permissive Mode
Often seen in advice “run in permissive mode to fix this”, permissive mode does not enforce the rules, but does report when it would normally stop an action. There is a danger however that while in permissive mode the file labels could be changed into a state which is unwanted:if a process tried to create a file but choses the wrong file label, permissive mode would allow this to happen, if you then return to encforcing mode, the new file would have the wrong label
cat /sys/fs/selinux/enforce – 1 indicates currently enforcing, 0 otherwise
Getenforce reports “enforcing if enforcing”
Setenforce 0 changes mode to permissive
Setenforce 1 changes mode to enforcing.

Permissive Mode
Usually best to fix things without leaving enforcing mode but people find this hard
You can create the file /.autorelabel then reboot - on the next boot the whole filesystem will be relabelled with the default labels for each file. Default labels may not be what you want and may override legitimate changes but labels will now be at least strictly correct. So if on boot default is enforcing, then at least the labels will work(the system should be bootable).

Security Concepts
Subjects - the thing performing an action. In linux these are processes, since processes are related to users, you can consider users subjects too.
Access - what access is permitted, such as read, write,append, delete,open,change
Object - the resource on which the action applies.
Subjects use access on Objects.

Domain
In SELinux, a label assgined to a process is also called a domain
$ ps -Zx | grep sshd
system_u:system_r:sshd_t:s0-s0:c0.c1023 2141 ? Ss 0:00 /usr/sbin/sshd
SELinuxUser:SELinuxrole:SELinuxType:SELInux MLS
So here the domain is System_u:System_r:sshd_t
For convenince, this is often shortened to type part, sshd_t

Type
A type is a label for an object, such as a file
ls -Z /etc/ssh/ssh_host_ecdsa_key
system_u:object_r:sshd_key_t:s0 /etc/ssh/ssh_host_ecdsa_key
When making deicions about an action, selinux considers the domain and the type but also the class, an obejct class is the description abuot what sort of object it is. For instance, a file, a direcotry, a soft link, etc. this allows fo rfine grained decision making.

SELinux: the processing
SELinux makes decisions from the AVC(access vector cache). If the subject/type/action pair is not in the AVC, it is loaded from the policy database first. -if the action is ok in the cache for a subject/type, then the action is allowed. If the action is not ok then an AVC: denied message is generated.

MLS
Sensitivity is the 4th field in a domain or type:system_u:system_r:sshd_t:s0-s0:c0.c1023
The fourth field has two concepts: sensitivity and category. Sensisitivity is equivelent to document classification like intternal,secret, top secret, etc represented by a number.
S0 is the least secret and higher nummbers represent more secrecy.
Category relates to concepts like department name or project titles. Again these are just numbers like c0 or c43 - unlike sensitivity c1 is not more or less than c0, just different.
MLS information can be written in a number of ways:
System_u:system_r:sshd_t:s0-s2:c0.c1023
Sensitivity of s0 to s2 and all categories from c0 to c1023
system_u:system_r:sshd_t:s0
• Sensitivity of s0 and a default categories from c0

system_u:system_r:sshd_t:s0:c5,c9,c44
• Sensitivity of s0 and membership of categories c5,c9, and c44.

MLS in practice - dominance
If a process has all the categories of the object(or more) , and a sensitivity equal to or more than the object, then it is said to dominate. Known as dom. If a process has some(at least 1) but not all of the categories, and a sensitivity equal to or less than the object, then it is said to be dominated. Known as domby.
Processes which ahve the same sensitivity and categories as the object is known aas eq(assume equal)
Process has categories which have nothing in common with the object it is said to be incompatible. Known as incomp
This is then used in an MLS policy. Such a policy might be that: read operations allowed if the domain ddominates the object, write operations are allowed if thre domain is dominated by the resources.

MLS - the last word
MLS policy is not currently used in SELinux by default. Type enforcement(using the other three fields of the label) is much more common, and much easier. However you expect to see the MLS fields appear wherever selinux is used, safe to ignore for now

SELinux Users
Domains have an SELinux user entry
Provides funcitonality which relates to something users cannot change themselves, each linux user is in reality also an SELinux user, based on a mapping
semanage login -l

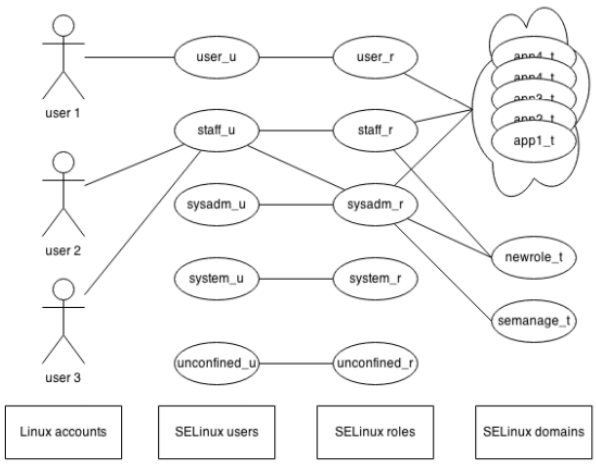
Login Name	SELinux User	MLS/MCS Range	Service
__default__	unconfined_u	s0-s0:c0.c1023	*
root	unconfined_u	s0-s0:c0.c1023	*
system_u	system_u	s0-s0:c0.c1023	*

Unconfined
Unconfined in SELinux indicates that SELinux will not enforce any rules in policing the interactions, in effect, if a user is unconfined, they operate largely as if SELinux is disabled.
Genreally if a user or process becomes unconfined then there should be no way back for the process to later become confiend
In the targeted mode, systems genreally use SELinux to enforce system-level concepts such as daemons - users are left unconfined.

Roles
Roles in SELinux are what actually carry permissions for users - roles get permissions, users get roles

```
# semanage users -l

SELinux User    ...        SELinux Roles

root                       staff_r sysadm_r system_r unconfined_r
staff_u                    staff_r sysadm_r system_r unconfined_r
system_u                   system_r unconfined_r
unconfined_u               system_r unconfined_r
user_u                     user_r
```

Default Roles
Roles and description
User_r - end user role meant to be assigned to users with interactive logon privileges to the system but without additiona role requirements
Staff_r - end user role meant to be assigned to users who also need to transition to other roles. The staff_r role by itself isnt mucch more privileged than user_r expect that it is allowed to do role tranistions
Systemadm_r - powerful role for generic system administration
System_r - system role meant for daemons and system services



The Apachje Daemon
$ ps auxZ | grep httpd
system_u:system_r:httpd_t:s0 /usr/sbin/httpd
When Apache wants to log errors, it writes it to /var/log/httpd/error_log
$ ls –Z /var/log/httpd/error_log
-rw-r--r--. system_u:object_r:httpd_log_t:s0 error_log

For http_t to access httpd_log_t, it needs a rule which allows this:
sesearch --allow --source httpd_t --target httpd_log_t -- file
allow httpd_t httpd_log_t : file { lock append open …} ;

SELinux ports
SELinux also ahs rules about what types can open whjich network ports, again for httpd, it would natrually want to use port 80
sesearch -s httpd_t -c tcp_socket -AC -p name_bind
allow httpd_t http_port_t : tcp_socket name_bind ;

semanage port -l | grep http_port_t
http_port_t tcp 80, 81, 443

Transitions
Processes can move from one type to another by executing a program of that new type - ofcourse there must be a rule which permits that transition.
Ffor example, the init process which starts all daemons has type init_t. You may wishj to run a command in /etc/init.d so that you end up running that daemon unconfined. In this case a special type exists, unconfined_service_t, which has no transitions back to a confined state.
To find the right type:
$ sesearch -T -s init_t | grep unconfined
type_transition init_t bin_t : process unconfined_service_t;
type_transition init_t usr_t : process unconfined_service_t;

SELinux Administration
The low level rules and tranistions are generally things never needing touched, but it is good to udnerstand the basics of how it works. In general terms all that is generally needed is - an ability to set labels to suit particular scenarios, an understanding of the error messages when things go wrong, some high level configuration via the booleans

Selinux usage
A system administrator, day to day selinux is generally much more high level and mechanical, to help essential SELinux offers, an automatic labelling service, commands to perform manual type changes, control over boolean activation of additional rules, a straight forward diagnostic tool with suggested changes, a simple tool to extend SELInux to new problems

Automatic laabelling
When a file is created it needs a type immediately. The average user needs SELinux support without needing any knowledge, by default its type is inherited from the parent. It is possible to create a type transtion based on a filename
$ filetrans_pattern(staff_t, user_home_dir_t, httpd_user_content_t, dir, "public_html")
This is best left to the policy writers rather than for everyday use. However it is useful

Warning!
Inherting the partent type only happens when a file is created, similarly, file transition patterns only happen at creation time. If a file is renamed(using mv) the current type stays as it is, this can lead to confusion if you accidenltally created a file, direcotry or whole directory tree initally in the wrong place

$ mkdir public_html
# ls -Z
… unconfined_u:object_r:httpd_user_content_t:s0 public_html
rmdir public_html
mkdir public_error
ls -Z … unconfined_u:object_r:user_home_t:s0 public_error
mv public_error public_html # ls -Z
… unconfined_u:object_r:user_home_t:s0 public_html

Automaticish labelling
One cannot expet the average user to write their own transition policy, instead uders have to accept the automatic type but can then relabel easily, to help adatabase of default types is maintained, these are not automatically assigned at creation time though, each user needs to run a command in order to get teh default type set - otherwise they get the inherited type or the name transition type.#
A database of default types is held. - at relabelling time the full pathname is used to look up the default type. Databases hold rules as regular expressoins
You can simply list all the rules:
$ semanage fcontext –l | grep passwd
/etc/passwd[-\+]? regular file system_u:object_r:passwd_file_t:s0
$ matchpathcon /etc/passwd
/etc/passwd system_u:object_r:passwd_file_t:s0
Auto RE-typing
So if you move the file and kknow you got the wrong type what do you do? For basic things, you can always reset a file to its default type with restorecon
$ matchpathcon /root/test
/root/test system_u:object_r:admin_home_t:s0
$ ls -Z /root/test
-rw-r--r--. root root unconfined_u:object_r:user_tmp_t:s0 /root/test
$ restorecon –v /root/test
restorecon reset /root/test context unconfined_u:object_r:user_tmp_t:s0->unconfined_u:object_r:admin_home_t:s0
$ ls -Z /root/test
-rw-r--r--. root root unconfined_u:object_r:admin_home_t:s0 /root/test

Auto RE-Typing a whole direcotry tree
Sometimes you might create directories and files and relaiase it has to be moved,
Simplely mv them as normal to the new location, then restorecon with -r(for recursive) and give the mathname to the top directory. Use carefully as you get the defaults, if a user deliberately changed a type and you relabel the whole of /home, you will overwite those changes, you can use -n if you are concerend, this ttells you what would change without actually changing anything
$ restorecon –rnv /home

Manual changes
Users who know what they are doing may want to change labels manually, this is done using chcon, note this changes only the specific file type, not the default type, if after manually changing a file you do restorecone the file changes back to the default!
$ ls -Z test
-rw-r--r--. root root unconfined_u:object_r:admin_home_t:s0 test
$ chcon -t bin_t test
$ ls -Z test
-rw-r--r--. root root unconfined_u:object_r:bin_t:s0 test
Parmanent manual changes
You can reprogram the restorecon type for things if you wish - that way restorecon will set the type to what you really want

So if you always want /root/test to be bin_t, you can do:
$ semanage fcontext -a -t bin_t "/root/test"
Do remember that the database is a list of regular expressions, so a default for a file”test.dat” would need to be “test\.dat”
• You can also use regular expressions to do complex matching
$ semanage fcontext -a -t bin_t "/root/(.*/)?test\.dat"

Using Booleans
Booleans allow rules to be switched on and off in groups - each group of rules linked to a boolean effectivly allow ro block a feature you may be inetersted in, if you can find a boolean to allow what you want to do, this is much easier than writing your own SELinux rules, getsebool allows you to read current boolean states, setsebool allows you to set current states.
List all booleans and their states with:
$ getsebool -a
…
httpd_enable_homedirs --> on
…

Setting booleans
Get the value fo a named boolean
Getsebool httpd_enable_homedirs
Http_enable_homedirs → off

Set the value fo a named Boolean(remembered till next reboot)
setsebool httpd_enable_homedirs 1
getsebool httpd_enable_homedirs
httpd_enable_homedirs --> on

Set the value of a named Boolean permanently
Setsebool -P httpd_enable_homedirs 1

But which boolean do you need?
From the theory you could work through the rules, however often you can get this sort of information from a manual, you can get a little commeting text using the following
$ semanage boolean -l | grep httpd_enable_homedir
httpd_enable_homedirs (on,on) Allow httpd to enable homedirs
There are also commands which analyse errors in the selinux log and propose solutions, and these may be booleans - in the remaining slides diagnostic approaches are discussed, and you will see that boolean changes are sometimes proposed automatically

Diagnostics
If something means an action is blocked by SELinux then it is logged, the log location is /var/log/audit/audit.log, the file can contain many issues some of which are not interesting, Lines which start with “type=AVC” aare the ones wihch are significant
A useful command is ausearch, which is a common tool to analyse log files

Ausearch -m avc -ts recent

This command files ABC errors in teh audit.log file which have occurrred within the last 10 minutes
type=AVC msg=audit(1490555793.938:152): avc: denied { read } for pid=1800 comm="httpd" name="httpd.conf" dev="dm-0" ino=26080829 scontext=system_u:system_r:httpd_t:s0 tcontext=system_u:object_r:named_conf_t:s0 tclass=file
$ date –d @1490555793.938 Sun 26 Mar 20:16:33 BST 2017

Deined{read} - action which was blocked,
Pid - thePID of the process affected,
Comm - the short name of the process running,
Name - the name of the target which was being accessed,
Dev - the device of the object.
Ino - the inode number of the object
The device dm-0 indicates lvm filesystem and you can find out more by doing:
$Dmsetup info /dev/dm-0
In this case dm-0 is the root filesystem /, so to find the file do
$Find / -xdev -inum 2608029 -> /etc/httpd/conf/httpd.conf
Scontext - the label of the process making the request.
Tcontext- the label of the target
Tclass - what was the target(in the case it is just a file)
It could be an error and you can find out:
ausearch -m avc -ts recent > file
[root@host-5-65 ~]# cat file | audit2why Was caused by: Missing type enforcement (TE) allow rule.
Often the error message is suggesting SELinux Booleans which if enabled would allow this action, and this generally can be very useful.
If it is not an error, you can actually add more rules to the database and permit this action
ausearch -m avc -ts recent > file
[root@host-5-65 ~]# cat file | audit2allow
#============= httpd_t ==============
allow httpd_t named_conf_t:file read;
However iun this case it is just a mislabelled file
restorecon -v /etc/httpd/conf/httpd.conf
 restorecon reset /etc/httpd/conf/httpd.conf context
system_u:object_r:named_conf_t:s0-
>system_u:object_r:httpd_config_t:s0

SELinux for systems
Eople are scared of SELinux - it is compelx sounding, good error messagesbut can be cryptic, good range of diagnostic tools but knowledge of them is poor, excellent layer of security but many still like running in permissive mode, try to work with it switched on and instead use unconstrained for processes which cause problems in SELinux: this at least secures some aspects of linux

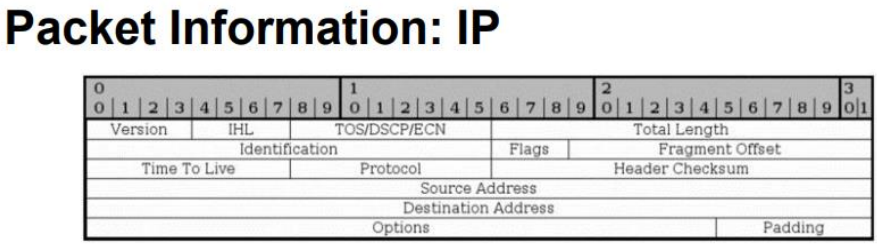
Essential Firewalls
Firewalls

Firewalls and the implementation of management policy plus proactive security methods. Management policy can include - network services allwoed to run and under what conditions, client access to external resources, quality of service, security and maintaining availability.

Corporate firewalls
Cisco ASA,Cisco router ACLs, linux router iptables, ASA syntax is strange, yet ASA solutions are popular!, ACLs are weak for policy implementations, iptables is powerful but uncommon.

Linux Firewalls
Security issues which linux firewalls can deal with:Denial of service, differences between external and internal users, hiding local only services, traffic management, virus and hacking protection, implementing management policy
Rules are needed in terms of packets arriving,leaving, and passing through.

Packet Information:IP

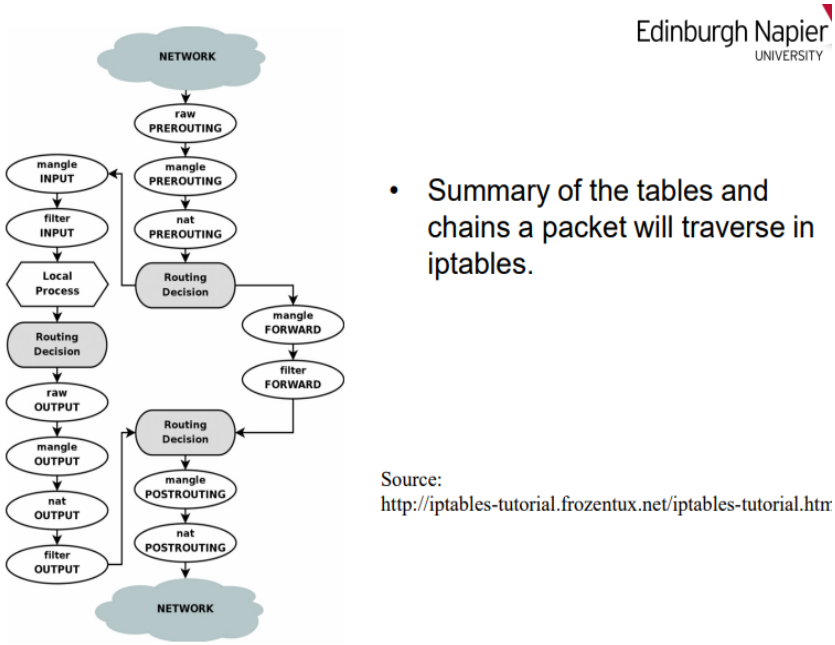


Decisions can be made about packets based on their IP header:packets from 10.0.0.1 are to be trusted, packets sent to 10.1.1.1 should be blocked, ICMP protocol packets should be blocked.
DEcisions can be made about TCP packets based on their TCP header: packets sent to port 80(http requests) should be allowed, packets sent from port 57890 should be blocked, SYN packets with FIN(an invalid combintion should be blocked.

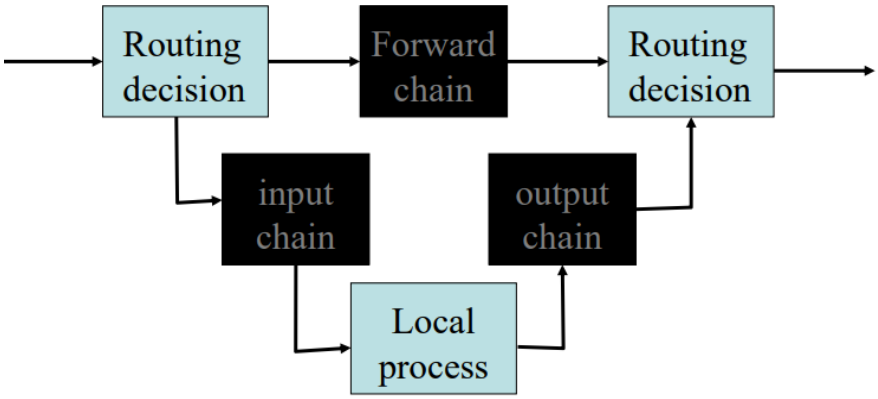
Iptables
Since kernel 2.4, the standard interfaces for firewall configuration is iptables. It implements its rule using many “tables” - filter - handles standard “firewall” things, NAT - rewriting of sourcce/destination IPs, Mangle - Specialised hacking of packet info, raw - low level modifications to packets
Most firewalls only ened to be involved with the filter table.

Chains
Within each table is a number of chains, think of each table as containing diferent types of rules you might want to use(like “filter” rules).
Thinkn of chains as defining when a packjet will have those rules applied to them. Chains are done in a particular order. Some packets only go to some chains and not others(depending on how the packet was made.

Chain Names
Some chain names you might see are
– PREROUTING – INPUT – OUTPUT – FORWARD – POSTROUTING



Filter tables chains: input, output,forward



Filter Table
In the filter table a packet will only go through 1 chain, packets created within the computer go through Output, packets which need to be routed from one eth device to another eth device go through forward, packets received by the computer for processing locally go through input

A Chain
Each chain is made up of 0 or more rules, a rule is a set of tests and an action, if the tests are all true then the action is performed, if a test is partially or completely false then the next rule is looked at instead, if all the rules are used up withouit an action taking place, then the chain policy is done instead(a default)

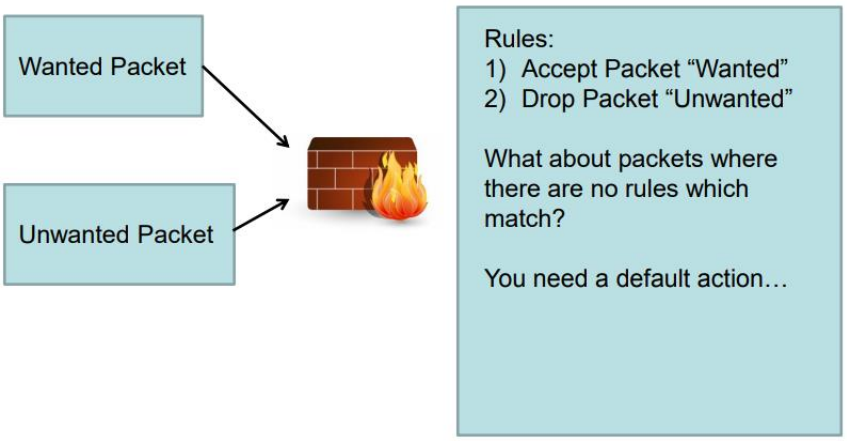
Tests
Each rule has 0 or more tests
There are many tests possible, such as: – Is this TCP? – Is this from 10.0.0.5? – Is this from port 22? – Is this going to port 23? – Is this going to ip 50.0.0.1? – Am I receiving packets

faster than 10 per second? – Which interface is it going out/in on? Remember you can combine these tests, e.g. TCP and from port 22.

Actions
If all the tests are true then the action is carried out. Some actions are terminating, meaning that once they are done no more rules are looked at. A few actions are nonterminating, this is useful if you want to say print a message if the test is true, but continue on with the next rule anyway.

Example actions are: – DROP – delete a packet immediately and terminate – ACCEPT – the packet is good and terminate – REJECT – delete the packet and terminate, but send back an ICMP message to the sender – LOG – print to syslog a message and move onto the next rule.
Some tests: • Is this TCP ? -p tcp • Is this from 10.0.0.5? -s 10.0.0.5 • Is this from port 22? --sport 22 • Is this going to port 23? --dport 23 • Is this going to ip 50.0.0.1? -d 50.0.0.1 • Is this going out on eth0? -o eth0 • Is this coming in from eth0? -i eth0

Default Policy



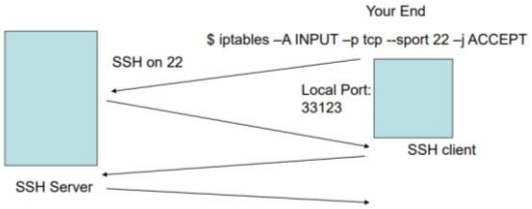
Default policy accept?
With default accept, all packets are accepted unless you have a rule to drop or reject them, you must have a rule for any bad packets you could receive.
Default policy:DROP?
With default ddrop, all packets are dropped unless you have a rule to accept them. You must have a rule for any good packets you could receive. More secure but easier to lock yourself out by accident

Setting the policy
$ iptables –P INPUT ACCEPT
$ iptables –P OUTPUT ACCEPT
$ iptables –P FORWARD DROP
This is a typical unsecured machine configuration. Typical machines only have 1 eth device, so dont forward. Otherwise all packets are allowed
Betterr to have input drop and have the accept rules but remember input drop without rules drops everything!
$ iptables –P INPUT DROP

Editing firewalls
Iptable does allow you to edit firewalls dynamically, however this is very problematic and difficult. instead i recommend putting all your rules in a file and running that file to change the firewall, this allows you to use your favourite editor to write the firewall. At the start of the file, delete all current firewall rules in each table using “-F”
$ touch firewall
$ chmod +x firewall
$ vi firewall
/sbin/iptables -F INPUT
/sbin/iptables -F OUTPUT
/sbin/iptables -F FORWARD
Set the default policies for the chains
/sbin/iptables -P INPUT
DROP /sbin/iptables -P OUTPUT ACCEPT
/sbin/iptables -P FORWARD DROP
To load these rules do $./firewall
However dont do that yet. The default is drop for input. Without more rules you will be kicked out of the server never to return, this is bad if the server is 5 minutes walk away, but if it is 5000 miles away you are in trouble!
This type of firewall is INGRESS ONLY. no rules for going out (OUTPUT/EGRESS) kind of like the XP firewall

SSH server example
Allow someone to SSH and TELNET into your machine
Standard stateless INGRESS FILTER RULES, then
iptables –t FILTER –A INPUT –p tcp --dport 22 –j ACCEPT
iptables –A INPUT –p tcp --dport 23 –j ACCEPT
When you ssh to elsewhere, you first packet goes out on the output chain to dport 22 on that remote machine, however the reply comes from dport 22 that remote machine to your computer via the input chain.

Sport or dport
Server end
$iptables -A INPUT -p tcp --dport 22 -j ACCEPT
$ iptables –A INPUT –p tcp --sport 22 –j ACCEPT



SSH Client Example
iptables –t FILTER –A INPUT –p tcp --sport 22 –j ACCEPT
So this rule intends to allow the reply(and all subsequent packets) to be accepted where it relates to someone on the local machine using ssh to connect to a remote machine. But a hacker from any machine could just send packets witha sport 22, and these would be

accepted by the firewall even though they are unwanted - this could allow an exploit to be used against our computer
We really need a way of knowing a packet is part of a TCP stream we actually want

Stateful rules
If you could find out what packets are the start of a stream and which packets are not, then you could have rules to control the first packet - if packet 1 then check to see if the rules say it is ok, if packet 2 then it must have been ok, otherwise packet 1 would have been blocked. Linux holds a state table of packets to allow such things to be monitored

You can add iptables rules to detect what state a packet is in. this feature used to be called “state”, but now is known as “conntrack” or connection tracking.
Tracking test available includes: – NEW – never seen the stream before – ESTABLISHED – traffic in both directions in the stream. – RELATED – associated with an established stream but is actually a different stream. For instance an established FTP connection may transfer each file using a new stream (active connections). Each new connection is RELATED. – INVALID – worrying and should probably be dropped!

/proc/net/nf_conntrack
ipv4 2 tcp 6 431978 ESTABLISHED src=146.176.164.219 dst=146.176.166.41 sport=56749 dport=22 ...
Here for instance a TCP stream has exchanged SYN, so is ESTABLISHED

ipv4 2 tcp 6 81 TIME_WAIT src=10.200.0.1 dst=10.200.0.19 sport=63040 dport=80 ...
Streams can end up in many states. Time_wait means the stream is clsoed, but the table remjembers closed connections for a time so that left over retransmissions can be handled properly.

Iptable rules based on network state
If a connection has been going a while
$ iptables –A INPUT –m conntrack --ctstate RELATED,ESTABLISHED –j ACCEPT
But a new incoming connection to the local ssh server needs to be allowed
DROP or REJECT
Drop just blocks the packets. Reject blocks and also sends an ICMP message to the sender saying it is blocked. DROP gives nothing away, and a hacker finds it difficult to get useful information about your network if packets are dropped. REject gives away information about your firewall ruleset.
With Drop and ip , where packets are lossy, some protocols may retransmit a packet which you have dropped thinking it has been lost. This means dealing with the same packet again

A goold rule: intranet traffic should be reject, internet traffic should be drop
Note yohu can only accept or drop in a policy but you can reject in a rule.

Example:tight ssh server example
Allows someone to ssh into your machine, provided their ip is 10.0.0.1
Standard stateful INGRESS FILTER RULES, then iptables –A INPUT –p tcp --dport 22 –s 10.0.0.1 –j ACCEPT • Allow someone to SSH into your machine, provided their IP is

between 10.0.0.1 and 10.0.0.255. iptables –A INPUT –p tcp --dport 22 –s 10.0.0.1/24 –j ACCEPT

Basic client machine
• Allow local machine to ssh only to 10.0.0.1 • You have been asked to use only the INPUT chain and leave OUTPUT as ACCEPT # Standard stateful INGRESS FILTER RULES, then BEFORE RELATED/ESTABLISHED iptables –A INPUT –p tcp --sport ssh ! –s 10.0.0.1 –j DROP

Add a rule to permit ping
Anyone can ping this machine
#add to theend of the file
iptables –A INPUT –p icmp --icmp-type echo-request –j ACCEPT
Rate limiting
You can do statistics on connections and use that in the rules.
The limit module allows you to make checks to see how times per second a particular rule is rule.
For instance, it is nice to handle PINGS but if you were asked to handle too many then perhaps it is ok to ignore some of those?
It uses the limit module, and the test is --limit and a rate, such as 3/second, 5/minute etcc.
Exceeding the rate is false and within the limit is true

Add a rulke to permit safe ping
Anyone can ping this machine but i will only respond if the ping requests are slower than 2 per second:
Add to the end of the file, but before RELATED/ESTABLISHED
iptables –A INPUT –p icmp --icmp-type echo-request -m limit --limit 2/second –j ACCEPT
Might help protect a system from a “ping of death”

Monitor safe ping
Anyone can ping this machine, but i will onyl respond if the ping requests are slower than 2 per second. Faster than that gets a logged message.
Add to the end of the file, but before RELATED/ESTABLISHED. iptables –A INPUT –p icmp --icmp-type echo-request -m limit --limit 2/second –j ACCEPT iptables –A INPUT –p icmp --icmp-type echo-request –j LOG

Anyone can ping this machine, but I will only respond if the ping requests are slower than 2 per second. Faster than that gets a logged message, but don’t log more than 10 per minute…
Add to the end of the file, before RELATED/ESTABLISHED
iptables –A INPUT –p icmp --icmp-type echo-request -m limit --limit 2/second –j ACCEPT
iptables –A INPUT –p icmp --icmp-type echo-request -m limit --limit 10/minute –j LOG
Impact of Policy DROP and ACCEPT
iptables -P INPUT ACCEPT
iptables –A INPUT –p icmp --icmp-type echo-request -m limit --limit 2/second –j ACCEPT
iptables –A INPUT –p icmp --icmp-type echo-request -m limit --limit 10/minute –j LOG
Impact of policy drop and accept

iptables –A INPUT –p icmp --icmp-type echo-request –j DROP

Basic client machine
Allows local machines to ssh and telnet out, using input policy of accept, leaving all other chains empty and accept
/sbin/iptables -P INPUT ACCEPT # iptables –A INPUT –m conntrack --ctstate NEW –j DROP iptables –A INPUT –p tcp --sport 22 –j ACCEPT iptables –A INPUT –p tcp --sport 23 –j ACCEPT iptables –A INPUT –p tcp –j DROP • …using INPUT policy of DROP: /sbin/iptables -P INPUT DROP # iptables –A INPUT –m conntrack --ctstate NEW –j DROP iptables –A INPUT –p tcp --sport 22 –j ACCEPT iptables –A INPUT –p tcp --sport 23 –j ACCEPT

Basic client machine • Allow local machine to ssh only to 10.0.0.1, • …but using INPUT policy of ACCEPT: iptables -P INPUT ACCEPT iptables –A INPUT –m conntrack --ctstate NEW –j DROP iptables –A INPUT –p tcp --sport ssh ! –s 10.0.0.1 –j DROP iptables –A INPUT –p tcp ! --sport ssh –j DROP • …but using INPUT policy of DROP: iptables -P INPUT DROP iptables –A INPUT –m conntrack --ctstate NEW –j DROP iptables –A INPUT –p tcp --sport ssh –s 10.0.0.1 –j ACCEPT

INPUT ACCEPT
To be clear, a policy of accept is seripously silly, good to learn on but wide open to attack, accept policy means you need rules to block. Anything you didnt think about is accepted. Drop policy means you need rules to allow. Anything you didnt think abuot is dropped. Security relies on a good policy and that must be DROP.

Subroutines
Complex rules in a single chain are difficult to manage, linuxzoo server firewall has 200 rules, in the same way as programs, firewall rules benefit from subroutines. Subroutines are created with -N, deleted with –X, and can be reached with a target of “-j subroutinename” and returned with a target of “-j RETURN”.

Monitor Monitor safe ping subroutine
Anyone can ping this machine, but I will only respond if the ping requests are slower than 2 per second. Faster than that gets a logged message, but don’t log more than 10 per minute… Use a SUBROUTINE
iptables –X ECHO iptables –N ECHO Iptables –A ECHO -m limit --limit 2/second –j ACCEPT iptables –A ECHO -m limit --limit 10/minute –j LOG iptables –A ECHO –j DROP iptables –A INPUT –p icmp --icmp-type echo-request –j ECHO

Extra tests
Many other simple tests, use man iptables to find more,
For example: – Stop apache surfing the web: iptables –A OUTPUT state --state NEW -p tcp --sport 80 -m owner --uid-owner=apache -j DROP – Specify port ranges like: --sport 137:139 – Specify ip masks like: -d 10.0.0.1/24

Basic Stateful FORWARDING
You are running your firewall machine with 2 network cards, eth0 and eth1
Eth0 connects to the internet, eth1 to the intranet

In this regard eth1 is a gateway for your local network.

Eth0 is 10.0.1.1/24, Eth1 is 10.0.2.254/24 • You have two servers in your intranet. – M1 is 10.0.2.1/24, running an ssh server – M2 is 10.0.2.2/24, running an http server • GW Firewall FORWARD would be:
iptables -F FORWARD iptables -P FORWARD DROP iptables –A FORWARD –m conntrack--ctstate RELATED,ESTABLISHED –j ACCEPT iptables –A FORWARD –m conntrack--ctstate NEW –p tcp –i eth0 --dport ssh –d 10.0.2.1 –j ACCEPT iptables –A FORWARD –m conntrack--ctstate NEW –p tcp –i eth0 --dport http –d 10.0.2.2 –j ACCEPT

Egress filtering
Up till now you have been using ACCECPT as the default policy for OUTPUT. However this is not as secure as having a drop policy. Drop as the policy in output is called egress filtering. Although easy to completely mess up it is no harder than input drop policy. It limits output packets to only those which you explicity define. It could help reduce hacking attempts and the spread of viruses.

Complete EGRESS example
Configure a nonrouting server firewall which runs telnet and http servers. Users on teh server can ssh out. Use EGRESS filterting
/sbin/iptables -F INPUT /sbin/iptables -F OUTPUT /sbin/iptables -F FORWARD /sbin/iptables -P INPUT DROP /sbin/iptables -P OUTPUT DROP /sbin/iptables -P FORWARD DROP iptables -A INPUT –m conntrack --ctstate \ RELATED,ESTABLISHED –j ACCEPT iptables –A INPUT –m conntrack --ctstate NEW \ –p tcp --dport telnet –j ACCEPT iptables –A INPUT –m conntrack --ctstate NEW \ –p tcp --dport http –j ACCEPT iptables -A OUTPUT –m conntrack --ctstate \ RELATED,ESTABLISHED –j ACCEPT iptables –A OUTPUT –m conntrack --ctstate NEW\ -p tcp --dport ssh -j ACCEPT

Domonstration of packet capture
To capture packets, use tcpdump, it has many options and filters. One of the most basic things to do is to learn to use -n
This means that traffic captures will use ip numbers, this is not the default
The default is to convert ip numbers to hostnames using dns lookups, however this generates network traffic and you may end up caputing dns lookups too. Can be confusing.

Basic capture
17:28:04.196267 IP 10.0.4.33.ssh > 10.200.0.1.20140: Flags [P.], seq 1710432:1710800, ack 26241, win 2863, options [nop,nop,TS val 572236 ecr 3126306861], length 368
Packet from 10.0.4.33 on port 22 to 10.200.0.1 port 20140
It has a push flag set in the header.
Seq n:m indicates that n is the seq number of this packet and m is the seq number of hte next packet
To help, n is the seq number minus the starting seq number, so after the first exchange the seq n number starts at 1, length is the payload length.

Port selection

Tcpdump only usually captures from one device at a time. Tcpdump -D lists the device 1.virbr0 2.nflog (Linux netfilter log (NFLOG) interface) 3.nfqueue (Linux netfilter queue (NFQUEUE) interface) 4.usbmon1 (USB bus number 1) 5.ens3 6.any (Pseudo-device that captures on all interfaces) 7.lo
• Default is option 1, which may not be what you want • You can say “tcpdump –i ens3” to choose a device.

Verbosity
You can add -v for more information on each packet, or -vv for more still, or -vvv for execessive informatoin
With 0v for instance you can view the ip information too
17:38:31.537153 IP (tos 0x0, ttl 63, id 33471, offset 0, flags [DF], proto TCP (6), length 60) 10.200.0.1.31597 > 10.0.4.33.telnet: Flags [S], cksum 0xf844 (correct), seq 2619062292, win 2920 0, options [mss 1460,sackOK,TS val 3126934187 ecr 0,nop,wscale 7], length 0
So line 1 is teh ip informatoin , while line 2 holds the TCP information

Filtering
Normally all traffic is captures, but this can be excessive. To cope tcpdump has a filter based on expressions, this is generally typed in after all the flags. Simple expressions include
– port 23 – tcp – udp – icmp – src 10.0.0.1 – dst port 23 – host 10.0.0.1 – And many more

Example: capture ssh traffic • You want to analyse ssh traffic from 10.200.0.1. tcpdump –nvi ens3 host 10.200.0.1 and port 23

Packet contents
-A flag shows packet contents in ASCII
You can use –X flag for hexademimal too.

Masks
In rules such as iptables -A INPUT -s 192.168.0.0/16 -j DROP the sourec iup mask can be specified as a quad dotted mask:iptables -A INPUT -s 192.168.0.0/255.255.0.0 -j DROP
The mask can be any sort of bitmask, not just a valid network vlsm. Block any host with a 0 in the last octet:iptables -A INPUT -s 0.0.0.0/0.0.0.255 -j DROP

Optimization using masks
Consider this problem – Write one or more rules which will block TCP port 80 traffic from 10.0.0.0 to 10.0.0.255 if the last octet is even. – Use only 1 rule and use bitmasks to perform the optimization
Even numbers have 0 is the LSB, to be in 10.0.0.0/24 the first 24 bits are important to prove it. So bits 25 to 31 are irrelevant(dont care)
Last octet must be 00000001, or simply 1
-s 10.0.0.0/255.255.255.1 –j DROP

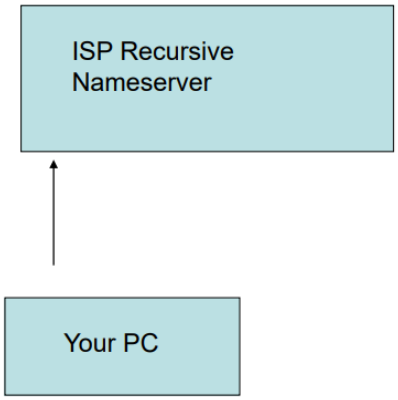
Networked Services
Domain Name Service - DNS
Translates between machine names and IP, two main types
-forward(domain to IP translation)
-reverse(ip to domain translation)

Terminology
Zone - a collection of hostnames and their IPs.
Nameserver - the server which responds to DNS queries. A question could be “give me the IP of grussell.org”.
Authoritative Nameserver - the server which has all the information for a zone stored locally.
Recursive Nameserver - if the nameserver is not authoritative for a zone, it is willing to go and ask other nameservers until it has asked an authoritative nameserver from the answer on your behalf. It then tells you the answer to your query.

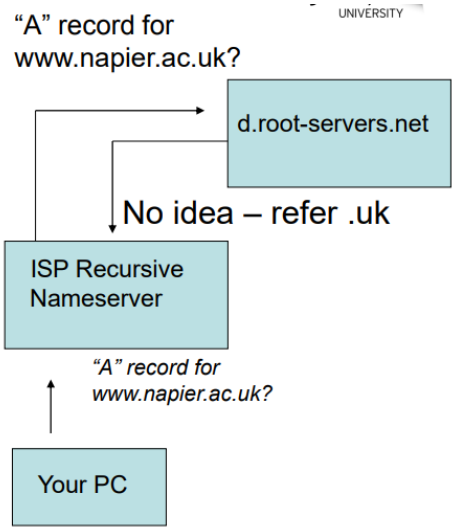
Resolver - the part of an OS which sends the DNS questions to nameservers. Its a library which other programs will use. For instance, “ping grussell.org” would ask the resolver to “resolve” grussell.org. It goes on to ask a nameserver for the answer.
Delegation - Sometimes a server does not know how to answer a query, but knows a server that can. The process of delegation effectively says that another server is delegated to answer your query, and you need to speak to them instead.
Resource Record(RR) - part of an answer to a query. An answer could be the IP for grusssell.org, but there are other resources records(eg for email delivery and delegation)

WHOIS
When you register a domain, you have to give information to the registrar.
This includes a contact name, address, and other contact details.
You also have to give at least 2 authoritative nameservers.
$whois napier.ac.uk
Registered For:
    Napier University
Servers:
    dns0.napier.ac.uk 146.176.1.5
    dns1.napier.ac.uk 146.176.2.5
Registrant Address:
    Napier University
    C&IT
    219 Colinton Road~Edinburgh

DNS Distributed Database
By way of an example, consider the following:
$ping www.napier.ac.uk
The resolver in linux is asked to find out the IP for www.napier.ac.uk . the resolver contacts its local recursive nameserver and send it a DNS query. The resource record needed to translate a domain name into an IP is known as an “A” record.
Step 1:the resolver contacts the local ISP nameserver.
It is not authoraitive. If it has been aked before it might have cached the answer however in this case it has not



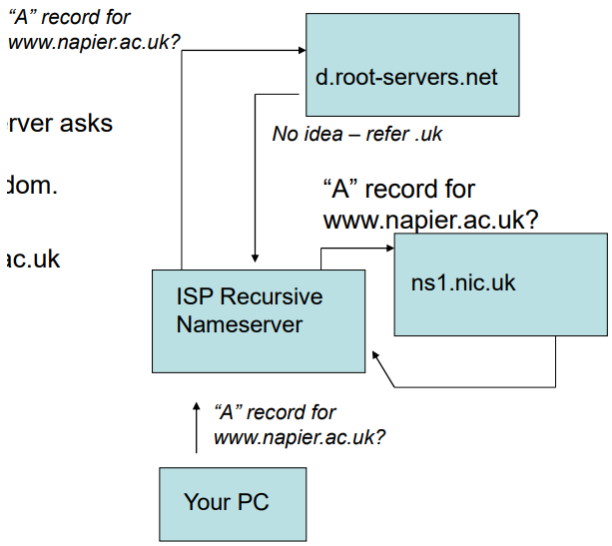
Step 2: ISP nameserver asks a root server… there are more than a dozen root servers. Their job is to direct your local nameserver to a nameserver which will help resolve the request. The root servers are pre-configured in the local nameserver, with names like:a.root-servers.net. b.root-servers.net. c.root-servers.net. D.root-servers.net.
It doenst know either but offers a referral to nameservers which can help



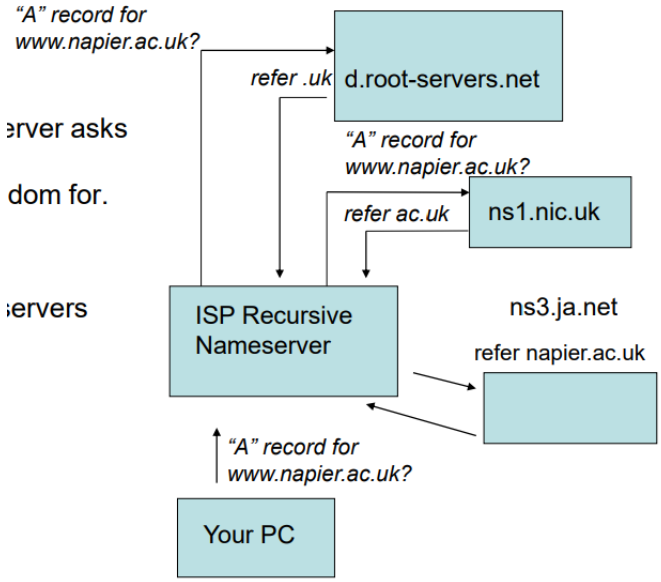
The root server refers to another set of servers which can answer the query:NS are nameserver resources records. Note that you also get the A records of the nameserver for free. This extra information is referred to as the glue recordsd. Without this we would have to look them up with a different query, so this saves time…

```
uk. 172800 IN NS ns1.nic.uk.
uk. 172800 IN NS ns2.nic.uk.
uk. 172800 IN NS ns3.nic.uk.
uk. 172800 IN NS ns4.nic.uk.
;; ADDITIONAL SECTION:
ns1.nic.uk. 172800 IN A 195.66.240.130
ns2.nic.uk. 172800 IN A 217.79.164.131
ns3.nic.uk. 172800 IN A 213.219.13.131
ns4.nic.uk. 172800 IN A 194.83.244.131
```

Step3 ISP nameserver asks one of the referral nameservers at random. Another referral, receives NS list for ac.uk Nameservers



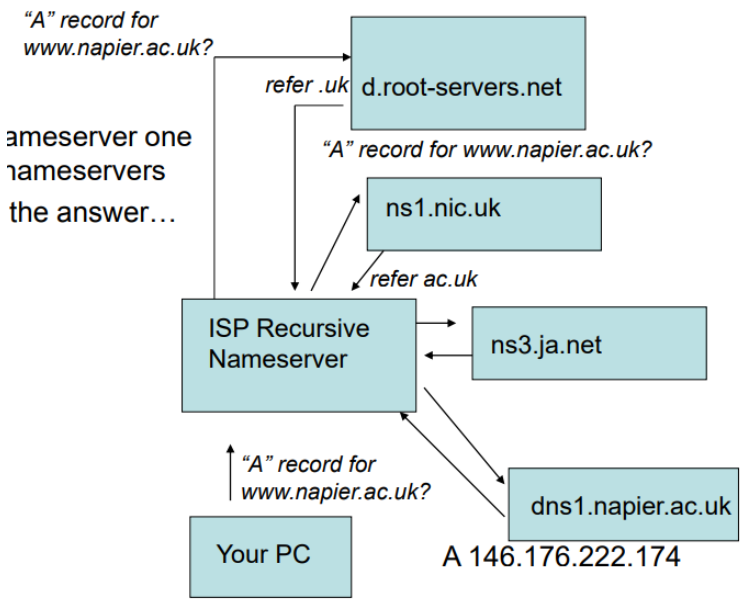
Step 4:isp nameserver ask one of the referral nameservers at random for. Another referral. Receive NS list for napier.ac.uk nameservers



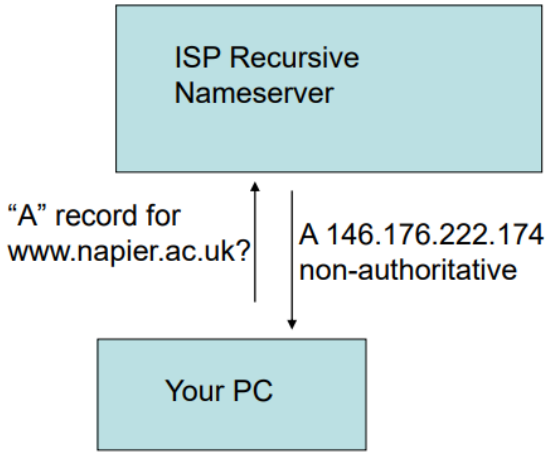
The ja.net nameserver responded with the authoritative nameservers for Napier. The last step is to ask one of the authoritative nameservers for www.napier.ac.uk

napier.ac.uk. 86400 IN NS dns1.napier.ac.uk.
napier.ac.uk. 86400 IN NS dns0.napier.ac.uk.
;; ADDITIONAL SECTION:
dns0.napier.ac.uk. 86400 IN A 146.176.1.5
dns1.napier.ac.uk. 86400 IN A 146.176.2.5

Step 5: ISP nameserver one of the napier nameservers, finally it gets the answer



The answer for Napier was AUTHORITATIVE. It is cached in the ISP. The cached version is returned to you. As it is cached, it is non authoritative.



Manual Lookups

If we are running our own DNS servers, or just having trouble making the resolver work, we can make our own queries using "dig"

$ whois napier.ac.uk
Domain servers in listed order:
dns0.napier.ac.uk 146.176.1.5
dns1.napier.ac.uk 146.176.2.5
> dig www.napier.ac.uk @dns0.napier.ac.uk
www.napier.ac.uk. 86400 IN A 146.176.222.174
;; AUTHORITY SECTION:
napier.ac.uk. 86400 IN NS dns0.napier.ac.uk.
napier.ac.uk. 86400 IN NS dns1.napier.ac.uk.
;; ADDITIONAL SECTION:
dns0.napier.ac.uk. 86400 IN A 146.176.1.5
dns1.napier.ac.uk. 86400 IN A 146.176.2.5

Reverse Lookup

Reverse is working out that given 146.176.222.174, the host is www.napier.ac.uk . a special domain name is used for IP to Domain Name translation. The domain is the IP in reverse, ending with IN-ADDR.ARPA
You need to take the first 3 elements of the IP first to find the right server, then query that server with the full IP. The resource record for reverse DNS is PTR.
> dig 222.176.146.IN-ADDR.ARPA … ;;
AUTHORITY SECTION:
222.176.146.IN-ADDR.ARPA. 86400 IN SOA dns0.napier.ac.uk. root.central.napier.ac.uk. 200808271 28800 7200 604800 86400
> dig 174.222.176.146.IN-ADDR.ARPA @dns0.napier.ac.uk -t any …
;; ANSWER SECTION:
174.222.176.146.IN-ADDR.ARPA. 86400 IN PTR www.napier.ac.uk.

Linuxx DNS
Resolver in Linux
You now know how DNS forward and reverse works. But you do not have to do all the repeated queries yourself! The resolver lkooks after all the lookups.
$ cat /etc/host.conf
Order hosts, bind
This is the schemes used to translate DNS.
Hosts - use the /etc/hosts file
Bind - use dns

/etc/hosts
This file is the simplest lookup. It is called “host” resolution. The file has lines like 127.0.0.1 localhost.localdomain localhost
It is IP , then hostname, then aliases for the host.
Its fast and simple, but only lists machines you edit yourself into the file. It is good for “kickstarting” finding key machines.

Cat /etc/resolv.conf
Search linuxzoo.net net
Nameserver 10.200.0.1

Not found in /etc/hosts ? check resolve.conf and use bind. Nameserver - who to ask(our nameserver), search - add these to the host if not found - Looking for www.linuxzoo? This search would try “www.linuxzoo”, “www.linuxzoo.linuxzoo.net”, “www.linuxzoo.net”. If still not found then fail. This is a convenience for users.

> dig www.linuxzoo.net @10.200.0.1
Ask your nameserver.

1. If your nameserver knows the answer, return it
2. If unknown, nameserver recurses(and stores the answer in cache).
3. If no one knows after recursion, return a failure.

If resolv.conf has multiple nameserver entries, each one is tried until all are tried or the answer is returned from one of them.

Your own nameserver

You might want to run your own nameserver if: you perform lookups frequently and want to cache the queries locally.you want to query the root servers directly without having to talk via a local nameserver. You want to add your own entries to DNS - you run your own domain and want to control your dns entries directly. You have local IPs and want to name them on your own network.

Nameserver daemons
The most popular is “named”. It is installed on the UML simulations. It is part of the bind9 distribution(ww.isc.org). It is popular, but other services are available.
Common systems are often targetting for hacking attacks. NAMED is often cited as a sevurity problem, needs to be patched often, it should be secured using addition security technology:can be run in a chroot, it can use sELinux if available, it can use both! In our labs we will use selinux security.

What is a chroot
Many linux service can be run “chroot”. This gives a new “/” directory just for that service. It contains only the minimum files and directories. The contents are only the minimmum files and direcotreies. The contents are (when possible) owned by a different user than the one who owns the service. The service should not be executed as root. Service runs as a user who does nothing else but run that service. Hack the service and you are stuck in a directory with little contents, you can change very little with a user which can do nothing if possible always run service in a chroot

SELinux Protection
In our experiments we will not be using a chroot for additional security. Instead we will use SELinux - this gives the kernel a set of rules which named must obey. Including what files can be opened and what sort of network connections can be made. Fortunately this is all preconfigured in fedora and thus completely invisible to us as administrators.

RNDC
Allows you to administer NAMED remotely
This happens with some security, RNCD uses a signed key to validate its security credentials. In aa non chroot solution this needs only to be stored in /etc/rndc.key . if you are using a chroot it must also be copied to /var/named/chroot/etc/rndc.key

Generate the key
In a normal installation, you should generate your own key.
In linuxzoo, you get the key generated automatically. This is a good thing as generating the key in linuxzoo turns out to be problematic. (So DONT)
Rndc-confgen -a -b 128 -r keyboard
-b 128 sets the bits to 128(fast but weak)
-r keyboard - random keys require “entropy”, normally done with /dev/random, bgut in UMLk this does not work well. This option asks you to type randomly for a while, and use your keyboard rhythhm to generate the random number. Only time you want to do this in linuxzoo is if you deleted the key file.

/etc/named.conf

Really two parts - options and zone
-options allow run time configurations and global default to be set. Zone entries allow us to set up a forward or reverse entry for a domain.

Master and Slave
There are two distinct types of zone:
Master - they have the zone defintions and you can edit that information if you wish.
Slave - they copy the one definition automatically from the master. Their copy is read only, so you cannot edit the records on a slave.
Slave nameservers are needed to give DNS higher reliability and redundancy. You edit on a master and the change is copied to all your slaves. Slave configuration is not considered further here.
The master is often called a primary, and any slaves called secondaries. But these names are badly abused, so stick with master/slave.
zone "." IN { type hint; file "named.ca"; };
This tells the daemon to use the root srevers listed in named.ca to resolve things not solved by other entries. This can be considered the “default”
options { directory "/var/named"; forward only; };
Nothing exciting in this part. Note that in linuxoo, DNS requests to the roots(or anywhere else) are intercepted my the linuxzoo fileway and redirected to 10.200.0.1 . this keeps the load on the root servers down, and makes it harder for people to use linuxzoo to hack the planet. allows allows the name service to falsify records - needed to make things work right in the UML’s
zone "localhost" IN { type master; file "localhost.zone"; allow-update { none; }; };
The file localhost.zone gives forward resolving for the domain “localhost”

zone "0.0.127.in-addr.arpa" IN { type master; file "named.local"; allow-update { none; }; };
The named.local file give reverse lookups for the 127.0.0.0/24 IP range.

Localhost.zone

```
$TTL 86400$
$ORIGIN localhost
@       1D IN SOA     @ root (
                       42 ; serial (d. adams)
                      3H ; refresh
                      15M ; retry
                      1W ; expiry
                      1D ) ; minimum
        1D IN NS @
        1D IN A 127.0.0.1
```

Named.local
$TTL 86400
@ IN SOA localhost. root.localhost. (1997022700 ; Serial 28800 ; Refresh 14400 ; Retry 3600000 ; Expire 86400) ; Minimum IN NS localhost. 1 IN PTR localhost.

Example : grussell.org, in IP 50.1.1.0/24
/etc/named.conf zone for this:

zone "grussell.org" IN { type master; file "grussell.zone"; allow-update { none; }; };
zone “1.1.50.in-addr.arpa" IN { type master; file “grussell.rev"; allow-update { none; }; };

> cat /var/named/grussell.zone
$TTL 86400 $ORIGIN grussell.org. @ 1D IN SOA ns1 admin.grussell.org. (2004101701 ; serial 3H ; refresh 15M ; retry 1W ; expiry 1D) ; minimum 1D IN NS ns1 1D IN A 50.1.1.1 www CNAME grussell.org. ns1 1D IN A 50.1.1.10

> cat /var/named/grussell.rev $TTL 86400 @ IN SOA ns1.grussell.org. admin.grussell.org. (1997022700 ; Serial 28800 ; Refresh 14400 ; Retry 3600000 ; Expire 86400) ; Minimum IN NS ns1.grussell.org. 1 IN PTR grussell.org. 10 IN PTR ns1.grussell.org.

MX (Mail Exchange) records

```
host1   IN MX 10 host1
        IN MX 20 backuphost
        IN MX 30 mx.easydns.com.
```

Priority goes to lowest number. No dot and end - add the origin.

Load Balancing

With 1 server providing a service, only that server can handle requests. Multiple servers can handle requests in parallel. But how can a single server name be made automatically share out requests to multiple servers? Load balancing does this.

```
host1   IN MX 10 smtp1
        IN MX 10 smtp2
        IN MX 10 smtp3
smtp1   IN A 10.0.0.1
smtp2   IN A 10.0.0.2
smtp3   IN A 10.0.0.3
```

Equal priority MX records are usually randomly utilised, selection mechanism is mail application dependent.

Email server balancing with A

```
host1          IN MX 10
smtp smtp      IN A 10.0.0.1
               IN A 10.0.0.2
               IN A 10.0.0.3
```

Chosen using rrset-order(default is random)
Make sure reverse of .1,.2,.3 -> smtp.domain.com

Server balancing with A

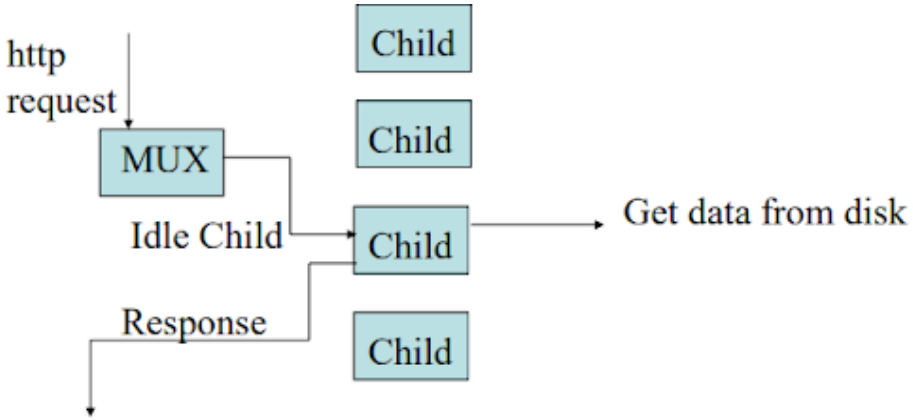
```
www     IN A 10.0.0.1
        IN A 10.0.0.2
        IN A 10.0.0.3
        ftp IN A 10.0.0.10
        ftp IN A 10.0.0.11
```

Chosen using rrset-order(default is random), the effect of caching needs to be considered, the distribution of load needs to be considered.

DNS record types
SOA - Start of authority, gives params for zone.
A6 - handle ipv6 references
NS - useful in delegation but basically ignored.
CNAME - an alias
HINFO - Hardware and OS being run
RP - who to send emails to.
TXT - useful text - for instance can be used to certify email server IP number for some spam detection software.

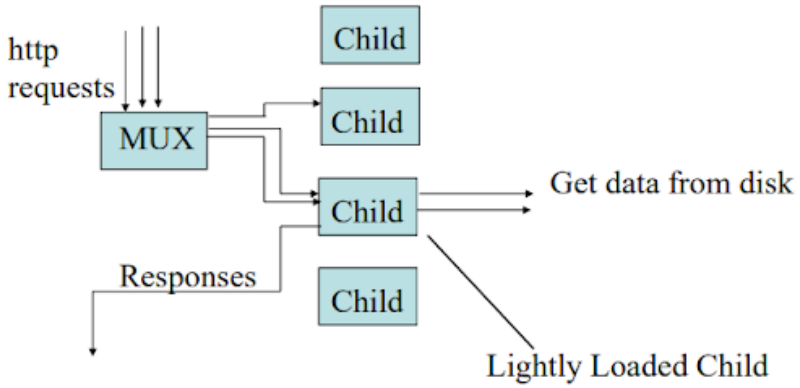
Apache
Server Architectures
Most designs of servers use:threaded model,forking model, asynchronous architecture.
Threaded model needs OS support to provide lightweight threads. Forking means each new request which arrives is handled by a whole process(tradtional apache way)
Asynchronous - one process handles everything with complex IO code. Good for fast processing of simple web pages
Hybrid - using more than one design simultaneously.

Tradtional forking model
Http request goes through the MUX which gives the task to a idle child, the chlid then gets the data from the disk and returns it



Forking Initial Settings
StartServers 8
MinSpareServers 5
MaxSpareServers 10
MaxClients 768
MaxConnectionsPerChild 4000

Worker threaded forking model
Http requests come in, the mux sends that to the childs that is doing the least amount of tasks, the response is then return.



Worker inital settings
StartServers 2

MaxRequestWorkers 150
MinSpareThreads 25
MaxSpareThreads 75
ThreadsPerChild 25
MaxConnectionsPerChild 0

Important files
/etc/httpd/conf/http.conf - the main conf file
/etc/httpd/conf.d/*.conf - where configuration changes go

in Centos7 this is in /etc/httpd/conf.d/userdir.conf.

Remember when changing configurations it is only reread on a server reload or restart .
Error and other details are logged by default in /var/log/httpd/ as access_log, error_log, as suexec.log

Reload or restart
Reload best option to use, apache checks configuration file and switches to it only if it contains no errors, if it has errors the old configuraiton keeps running. Allows reconfiguring a server with no downtime. Restart shuts down the server
Error log in /var/log/http/error_log) or syslog (/var/log/messages)
Systemctl start|stop|reload|restart|status httpd.service
Can check for errors using httpd -t

Mimic a browser
Telnet linuxzoo.net 80 HEAD / http/1.1 Host: linuxzoo
HTTP/1.1 200 OK Date: Wed, 15 Oct 2014 10:31:49 GMT Server: Apache/2.4.6 (CentOS) PHP/5.4.16 SVN/1.7.14 Content-length: 6876 Last-Modified: Tue, 14 Oct 2014 15:40:15 GMT Content-Type: text/html; charset=UTF-8

In http1.1 we canuse virtual hosts. Allows multiple hosts to share single server, each host has different name, name of host you wantto asnwer a query is given as part of page request. Only supposeted in http 1.1 and beyond.
VirtualHosts
<virtualhost *:80>
Sharing of single IP to provide multiple hostnames supported in apache, part of the conf file which handles tihs is called <VirtualHost> m each part holds list of hostnames it can handle, first host foud in the file is always considered the default, so if no virtualhost section matches the first block is done instead.
ServerAdmin me@grussell.org
DocumentRoot /home/gordon/public_html
ServerName grussell.org
ServerAlias www.grussell.org grussell.org.uk
ErrorLog logs/gr-error_log
CustomLog logs/gr-access_log combined
</virtualhost>

Public_html

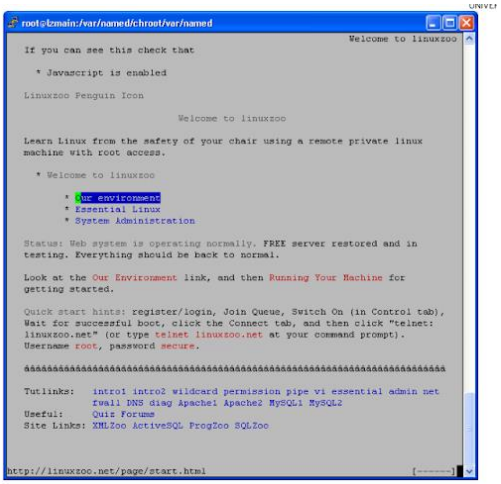
Where apache runs on a server used by many different servers, it would be useful for each user to be able to build their own webpages which the server could serve. Virtualhost configuraiton takes only a single document root, and each user has their own directories in /home, could make the root .home , however all of the files in /home become accessible Apache support web pages appearing in a users home directory, under the subdirectory public_html

Public_html access
Urls of the form - http://linuxzoo.net/~gordon/file.html
Refer to – /home/gordon/public_html/file.html
Feature must first be switched on in httpd.conf, to active it find the line User disble and either delete it or put # (comment) in front.
Then find the following line and delete the '#' character.UserDir public_html . remember to reload the server.

Linuxzoo tutorials
Each time you book a linuxzoo machine, you will likely get a different ip and hostname. Each time you come in check your hostname with "hostname".
$hostname host-5-5.linuxzoo.net.
In this example, virtual hosts vm-5-5.linuxzoo.net, as well as host-5-5 and web-5-5 will be proxied to your machine. • Warning: If the server on which your virtual machine fails, you will be moved to a different machine and a different IP. You need to check your hostname when you boot!

Web access from the prompt
Prompt is fast and convenient for admin purposes, but when you are debugging http sometimes "telnet" is not sufficient. Few other tools you can use at the prompt:
Elinks,lwp-request,wget
No simple replacement for actually using a real browser ot chekc your pages.

Elinks http://linuxzoo.net



Copy http to your directory
Lwp-request http://linuxzoo.net > file.html
Data is obtained and then printed to the screen, in case that is redirected to file.hml
Wget http://linuxzoo.net

```
$ wget http://linuxzoo.net
--19:20:11--  http://linuxzoo.net/
Resolving linuxzoo.net... 146.176.166.1
Connecting to linuxzoo.net|146.176.166.1|:80... connected.
HTTP request sent, awaiting response... 200 OK
Length: 4785 (4.7K) [text/html]
Saving to: `index.html'
100%[=========================================>] 4,785       --.-K/s   in 0s
19:20:11 (304 MB/s) - `index.html' saved [4785/4785]
```

SELinux and Apache

SELinux secures apache, and SELinux security files in public_html is b default quite strong

Check if selinux allows files to be published from public_html by

- getsebool httpd_read_user_content , if this is 0 then publishing files is forbidden.

Set SELiux tow allow public_html publishing using:

- Setsebool -P httpd_read_user_content 1
- May take 20 or more seconds, be patient, setting will be forgotten ifyou get a new imagein the linuxzoo interface.

SELinux requires the file security(shown by ls -Z) to be:

- Unconfined_u:object_r:httpd_user_content_t:s0
- However this should happen automatically provided you create files in public_html, an set the type of say filename.html(remmeber you should not have to) by using
- Chon -t httpd_user_content_t filename.html

URL Parts

• URL:

– http://linuxzoo.net/mystuff/file.html

• This is made up of

– Protocol: “http”

– Hostname: “linuxzoo.net”

– Path: “/mystuff/file.html”

• In URL rewriting, you often only get to operate on part of a URL in any one instruction, so take care you know which bit you want to use!

Basic for of this rule is:
RewriteRule URL-PATH New-PATH
• For instance, you have moved /dvd1.iso into a subdirectory /dvd…
RewriteRule /dvd1.iso /dvd/dvd1.iso
• The first parameter is a Regular Expression.
• This is an “internal” rewrite. In this case only Apache knows the changed location, and from the browser’s point of view the file appears to still be /dvd1.iso.

. Any single Character [chars] One of the characters in chars [^chars] None of the characters in chars

Quantifiers: ? 0 or 1 of the preceding text * 0 or N of the preceding text + 1 or N of the preceding text • Grouping (text) A text group – Can mark the border of an alternative or for RHS reference as $N

Anchors: ^ Start of the URL $ End of the URL • Escaping \char Allows you to use a character as the “char”. For instance, \^ is the ^ character and not the start of the URL.
You have directories “/android/1.txt”, “/android/2.txt”, and “/gordon/android/here/3.txt”. • Your manager has told you for marketing reasons you need to start using “droid” rather than “android”, so you rename the 2 directories. – But you want the old URLs to still work without having to keep copies of the old directories. RewriteRule ^(.*)/android/(.*)$ $1/droid/$2
At the end of the RewriteRule can be a number of flags. • The Flags are listed in [brackets], eg [F,G] for flags F and G. • These change or enhance the behaviour of the match.
• R or R=code - This sends the browser the new URL as an external REDIRECTION. The code can be the type or redirection, such as 302 for MOVED TEMPORARILY (the default). • F - Send back FORBIDDEN. • G - Send back GONE • C - Chained… if the test is false skip the next rule too. • L - Last – do not look at any more rules. • NC - case insensitive.
• If the URL starts /drgordon/, rewrite /drgordon/ to /lordgordon/. • In addition, if the URL did have /drgordon/ in it, replace any use of “hi.txt” with “greetings.txt”. RewriteRule ^/drgordon(.*) /lordgordon$1 [C] RewriteRule ^(.*)/hi\.txt$ $1/greetings.txt [L] • If rule 1 matches then rule 2 is checked • If rule 2 also matches do not do any other rules (e.g. rule 3 onwards). • If rule 1 does not match then go onto rule 3+.
Here are some of the important variables: – REMOTE_ADDR – REQUEST_METHOD – HTTP_HOST – REQUEST_URI (e.g. a path such as /index.html) (Its URI not URL) – REQUEST_FILENAME (e.g. /home/gordon/…) • You use these by using a percent sign, then enclosing the variable name with curly brackets, such as %{REMOTE_ADDR}. • There are many variables available for a variety of purposes
RewriteCond can also take flags in the same way as RewriteRule. • There are only 2 flags: – NC – case insensitive – OR – or the Conds together. • Normally all conditions have to be true before the RewriteRule is done, with OR the RewriteRule is done if ANY condition is true.

Example 1:
• If 10.20.0.5 tries to view any file, redirect the page reference to /gordon/bye.html.
RewriteCond %{REMOTE_ADDR} ^10\.20\.0\.5$
RewriteRule .* /gordon/bye.html [L]

Example 2:
• If 10.20.0.5 tries to view /gordon/index.html, redirect the page reference to /gordon/bye.html.

```
RewriteCond %{REMOTE_ADDR} ^10\.20\.0\.5$
RewriteRule ^/gordon/index\.html$ /gordon/bye.html [L]
```

Example 3:
• My VirtualHost has grussell.org, www.grussell.org, and www.grussell.org.uk.
• Rewrite all requests to grussell.org.

```
RewriteEngine on
RewriteCond %{HTTP_HOST} !^grussell\.org$
RewriteRule ^(.*)$ http://grussell.org$1 [L,R]
```

Example 4:
• Rewrite *.grussell.org to grussell.org, and *.grussell.org.uk to grussell.org.uk.

```
RewriteEngine on
RewriteCond %{HTTP_HOST} ^.+grussell\.org$
RewriteRule ^(.*)$ http://grussell.org$1 [L,R]
RewriteCond %{HTTP_HOST} ^.+grussell\.org\.uk$
RewriteRule ^(.*)$ http://grussell.org.uk$1 [L,R]
```

Example 5:
• If the request is for host “try.com”, and the path is “/here.dat”, then redirect externally to http://there.org.

```
RewriteEngine on
RewriteCond %{HTTP_HOST} ^try\.com$
RewriteCond %{REQUEST_URI} ^/here\.dat$
RewriteRule ^(.*)$ http://there.com [L,R]
```

Networked Services partical howto

It is only the best (i.e. Highest mark) for each question which is used to calculate your score. If it is not working think about all the things needed to make it work, and check them one at a time. • If it is a service problem, feel free to look at the error logs (rather than telepathy). • If it is hopeless reboot with a fresh image and try again in that question (really hopeless). • If things are just weird ask for help. I will help you if the problem is not something I am actually testing (e.g. you cannot start apache because of a known apache bug).

Do the questions in the order shown. • If you jump about between questions you will become confused. • YOU MUST DO THE CHECKS IN THE ORDER SHOWN. • If you configure your firewall, you may find you have blocked other services. Next trying apache or DNS will leave you doomed without resetting the firewall – Firewall reset: $ systemctl restart iptables

Apache Basic Authentication
Might want simple suernames and passwords to control access parts of a website, many appraoches. Easiest way to use basic authentication. Asked the browser to askj you fora username and password for accessing protected pages. Username and password is sent as clear text for every page request made by the browser. Https does get around thiss issue slightly, while it is still in plain text in http the whole request is encryupted with https

.htaaccess
Best way to control basic authentication is via an .htaccess file in the directory to protect. To allow the <directory> definition which includes the directory to be protected must have allowoverride authconfig. Essentially changes it so that one user can change the local access for people without editing the global configuration.

Building a password file
Have to create a file with usernames and passwords. Good idea if this file is not one which sommeone can access via a URL. Good idea to have it be one above the place it is supposed to be protecting! If there is any file in apache that begins with ht it cannot be downloaded! So if you wanted to store the password file in the document it is protecting you can call it htpasswd and apache will not show this to any users.
Htpasswd -c /homme/gordon/password andrew (type in password)
-c is only the first time running the command, creates the file too, miss out the -c after the first run or it wipes it!

.htaccess
AuthType Basic
AuthName "Restricted Files"
AuthUserFile /home/gordon/password
Require user andrew

Authtype digest - another option, requests the passwords in an encrypted format. It is not as widely supported as basic.

The password file
Password file created is jjust a text file, does not scale well! As more users are added the file gets bigger. On every page requyest the file has to be parsed again. Other formats available using hashed files(either db or dbm). Faster to access but more coplex to manage.

Any valid user
Require user andrew can be changed to require valid-user, in this way any user in the password file can access the directory.

Groups
Can use groups for protection, create plain text file withthe following format:
groupname:user1 user2 user3, if users gordon and andrew exists you want themnto be known as group staff, staff:gordon andrew
Add to .htaccess
AuthType Basic

AuthName "By Invitation Only"
AuthUserFile /home/gordon/password
AuthGroupFile /home/gordon/groups
Require group staff

Basic auth problems
Passwords in the clear, every request needs the password file lookup, large numbers of users are difficult to manage, not a good idea for commerical systems. However even with all these users recognise it and understand it!

Control by IP
.htaccess can offer more control than just basic authentication. Can also restrict access to directories by IP. need to use variations of the require directive
– Require user gordon – Require group staff – Require all granted – Require all denied – Require ip 10.0.0.1 – Require host linuxzoo.net – Require not ip 10 172.16 192.168.200
Multiple require statement can be combined together - default is that any can be true to allow access.

Example
Stop 10.0.0.1 accessing a directory… • Edit the .htaccess in that directory: Require not ip 10.0.0.1
By default the rules are combined together as any. So
Require ip 10.0.0.1
Require ip 10.0.0.2
Is actually
<RequireAny>
Require ip 10.0.0.1
Require ip 10.0.0.2
</RequireAny>
This may make statements easier to understand

Cbombining rules- aother methods
Requireany is basically a list of ors, if the first thing is true or the second thing is true
Other directives exist
– RequireNone: None of the directives must succeed
– RequireAll: All of the directives must succees
So to stop 10.0.0.1 and .2…
<RequireNone>
Require ip 10.0.0.1
Require ip 10.0.0.2
</RequireNone>

Given a basic authentication definition, allow user gordon, or group magic, or anyone from IP 10.0.0.50:
<RequireAny>
Require ip 10.0.0.50
Require group magic

```
Require user gordon
</RequireAny>
```

Given a basic authentication definition, allow user gordon from 10.0.0.1, or user jim from IP 10.0.0.50:

```
<RequireAny>
	<RequireAll>
		Require ip 10.0.0.50
		Require user jim
	</RequireAll>
	<RequireAll>
		Require ip 10.0.0.1
		Require user gordon
	</RequireAll>
</RequireAny>
```

Question 1

• The following is an .htaccess file of a fictitious student on a student's web account.

```
Auth Type Basic
AuthTitle "Password Required"
AuthUserFile home/jim/.www-password
Required user jim
```

• Spot 4 errors

AuthType not Auth Type
AuthName not AuthTitle
/home/jim not home/jim
Require user jim not Required user jim

Question 2

• The following is an .htaccess file of a fictitious student on a student's web account.

```
AuthType Basic
AuthName "Password Required"
AuthUserFile /home/09006754/.www-password
Require user server_admin
```

• Provide the code to change the password for Server_admin.

Htpasswd /home/09006754/.www-password server_admin

Apache Log Analysis

Logs

Apache produces two types of log files

Error logs, access logs

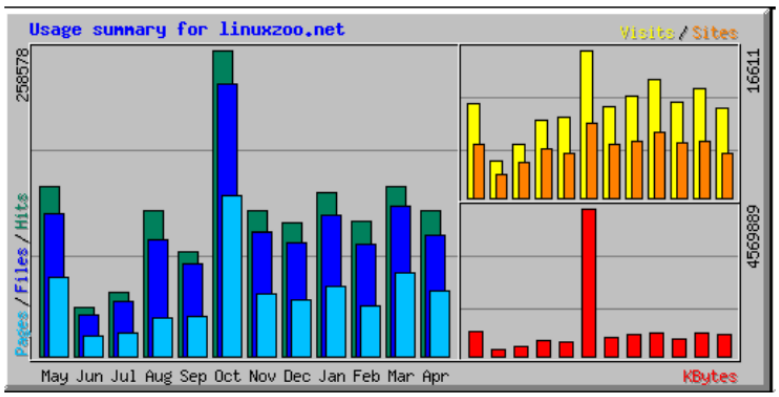
Error logs for debugging, access logs for monitoring how website is being used.

Where are the logs
Normally they go to /var/log/httpd/access_log and error_log
In a virtual host we set them to what we liked:

```
<virtualhost>
        ErrorLog logs/gr-error_log
        CustomLog logs/gr-access_log combined
</virtualhost>
```

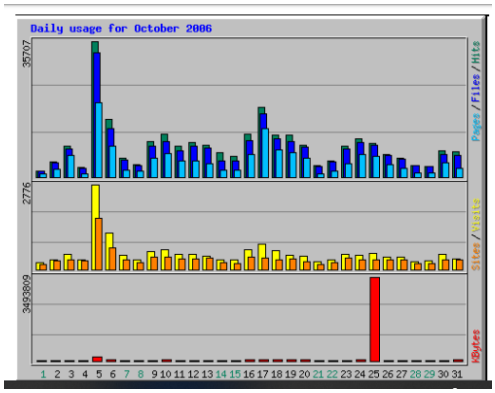
Logging in /var/log/http access file
Normally used log format is called combined
Contains signification amounts of information about each page request
Specifically, the log format is:
%h %l %u %t %r %>s %b Referrer UserAgent
H- ip of the client
I - useless ident info
u - username in basic authentication
t- time of request
r-the request itself
S-the response code(eg 200 forsuccessful request)
B-size of the reponse page
Referrer - link used to get there(maybe like google or something:?)
User Agent - identification info of the browser

Analysing the log
The log is useful in itself for checking proper function of the server. Traffic analysis also valuable, one of the best free ones is webaliser



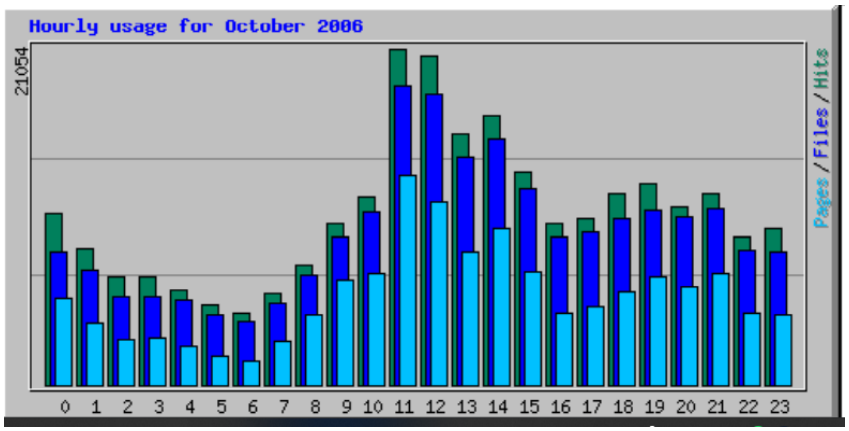
Analysis of this chart
Summer is quiet for linuxzoo, students are enthusatic in october, after that it settles down to “kept busy”

Per day activity – October



• I wonder which day was the first tutorial? • Look at the 7 day oscillations. This is common in many web sites. • Who stole all my web site data on the 25th?

Hour analysis – October



Peak learning time is 11am, students here seem to like 9am-4pm, american students produce another bump later at night

Users

#	Hits		Files		KBytes		Visits		Hostname
1	25319	9.79%	24605	10.69%	94474	2.07%	1051	6.33%	gtw-12.nhs.uk
2	16906	6.54%	16906	7.35%	14933	0.33%	1	0.01%	dsl081-020-069.nyc1.dsl.speakeasy.net
3	6400	2.48%	6381	2.77%	32857	0.72%	122	0.73%	200.182.252.4
4	6005	2.32%	5887	2.56%	24881	0.54%	245	1.47%	200.217.233.139
5	4903	1.90%	2230	0.97%	8144	0.18%	39	0.23%	mvx-200-196-55-148.mundivox.com
6	4506	1.74%	4506	1.96%	6580	0.14%	4	0.02%	200-221-128-3.corp.uolinc.com
7	3679	1.42%	9	0.00%	20	0.00%	2	0.01%	59.145.136.1
8	2728	1.06%	2726	1.18%	10462	0.23%	47	0.28%	59.163.124.54.static.vsnl.net.in
9	2690	1.04%	2590	1.13%	4733	0.10%	103	0.62%	58.68.28.66
10	2647	1.02%	2645	1.15%	11453	0.25%	193	1.16%	glenlivet.spc.eeng.liv.ac.uk
11	2479	0.96%	2455	1.07%	11568	0.25%	69	0.42%	mx2.queirozgalvao.com
12	2381	0.92%	2376	1.03%	50698	1.11%	1	0.01%	200.4.171.30
13	2156	0.83%	270	0.12%	2494	0.05%	6	0.04%	186.112-84-212.staticip.namesco.net
14	1657	0.64%	1657	0.72%	5821	0.13%	8	0.05%	80-41-249-174.dynamic.dsl.as9105.com
15	1606	0.62%	1601	0.70%	7577	0.17%	42	0.25%	146.176.242.35
16	1479	0.57%	347	0.15%	2545	0.06%	6	0.04%	r200-40-197-174.static.adinet.com.uy
17	1478	0.57%	1478	0.64%	0	0.00%	1476	8.89%	host.avidnetwork.com
18	1323	0.51%	1056	0.46%	12408	0.27%	3	0.02%	80-192-78-217.cable.ubr13.edin.blueyonder.co.uk
19	1157	0.45%	1155	0.50%	5840	0.13%	42	0.25%	146.176.242.54

Referrer Info

Top 30 of 831 Total Referrers			
#	Hits		Referrer
1	45948	17.77%	- (Direct Request)
2	1774	0.69%	http://www.google.com/search
3	425	0.16%	http://mail.google.com/mail/
4	343	0.13%	http://www.dicas-l.com.br/dicas-l/20061005.php
5	323	0.12%	http://www.google.co.uk/search
6	182	0.07%	http://www.google.ca/search
7	151	0.06%	http://en.wikipedia.org/wiki/System_administrator
8	142	0.05%	http://www.google.co.in/search
9	140	0.05%	http://www.ligeirinhorj.blogger.com.br/
10	135	0.05%	http://sqlzoo.net/
11	113	0.04%	http://146.176.165.229/my-netlab-s.cgi
12	99	0.04%	http://www.stumbleupon.com/refer.php
13	97	0.04%	http://www.google.com/linux
14	92	0.04%	http://www.google.com.au/search
15	79	0.03%	http://grussell.org/
16	78	0.03%	http://www.dicas-l.com.br/
17	73	0.03%	http://www.google.de/search
18	66	0.03%	http://www.google.nl/search
19	55	0.02%	http://www.grussell.org/

What search terms?

Top 20 of 1160 Total Search Strings			
#	Hits		Search String
1	176	8.17%	umask 022
2	68	3.16%	linuxzoo
3	46	2.13%	unix file structure
4	45	2.09%	apache2 authentication
5	31	1.44%	rndc key
6	30	1.39%	apache basic authentication
7	27	1.25%	apache public_html
8	20	0.93%	generate rndc key
9	18	0.84%	linux zoo
10	18	0.84%	named service
11	16	0.74%	iptables -f
12	16	0.74%	mysql replicate
13	16	0.74%	symbolic link example
14	15	0.70%	apache2 basic authentication
15	15	0.70%	setting umask
16	14	0.65%	apache basic auth
17	13	0.60%	named forward
18	12	0.56%	rndc key

Google Analytics

Another approach to web logging is to use javascript embedded in each web page, this does away with the need to asscess the web log, good if you dont have access, does mean that you only get logs where jkavascript is switched on, eachb page is slowed by having extra stuff on it, little mjore complex

Logging Summary • What is best? • I have used both and have mixed feelings… • Things to consider – Convenience – Reliability – Availability – Performance – Cost – Privacy – Complexity

Question 1 • Below is a line from a webserver logfile: 157.55.18.25 - - [31/Aug/2011:12:48:04 +0100] "GET /robots.txt HTTP/1.1" 200 48 "-" "Mozilla/5.0 (compatible; bingbot/2.0; +http://www.bing.com/bingbot.htm)" • What kind of request was this? Was this a successful request (i.e. was a document found)?
As seen after the section about date and time it is a Get request as can be seen by GET /robot.txt,as can be seen at the end bingbot was used so it is bing crawling over the site so that it can index it, it uses the robots.txtas a way to control which pages that should be crawed on and those which should not!, 200 so it was successful,

Apache security Discussion
Security
Hackers often consider a web server a good hacking target, should be very careful how apache is configured. Need ot be careful how apache is configured. Main problem is CGI scripts, CGI is a program which runs when you view a page. it s output is sent back to the users browser. As it is an active process it can do permanent things to your server.

Simple CGI:who.cgi

```
#!/bin/sh
echo 'Content-Type: text/html; charset=ISO-8859-1'
echo
echo '<body><pre>'
whoami
env
echo '</pre></body>'
```

http://servername/who.cgi

apache SERVER_SIGNATURE=Apache/2.0.51 (Fedora) Server at servername Port 80
UNIQUE_ID=umn4CZKwogYAADNFYkcAAAAI HTTP_KEEP_ALIVE=300
HTTP_USER_AGENT=Mozilla/5.0 (Windows; U; Windows NT 5.1; enGB; rv:1.9.0.1)

Gecko/2008070208 Firefox/3.0.1 SERVER_PORT=80 HTTP_HOST=servername DOCUMENT_ROOT=/home/gordon/public_html

Issues
This cgi proggram only prints however it could be used to delete things, transfer data, copy passwords etc. a hacker is rarely wanting distruction, they want access! This requires either transferring hacking programs to the server or copying files from the server(/etc/passwd)

Ideas
Make sure apache runs as a user just for the server, the user apache is commonly used hejre, in the httpd.conf, make sure there is user apache, group apache.
Hide the apache version number - might be userful if a hacker is searching for a buggy apache version, in httpd.conf ServerSignature Off,ServerTokens Prod
Dont allow apache to ever give pages from /

<Directory />
Require all denied
</Directory>
Do you really need directory browsing?
Options -Indexes
The apache user should not own its conf files
$ chown -R root:apache /etc/httpd $ chmod -R u=rwx,g=r,o-rwx /etc/httpd
Do not allow apache to surf the web:
$ iptables -A OUTPUT -m owner --uid-owner apache -m state --state NEW -j DROP

SMTP
Email sent between source and destination using a simple protocol. SMTP is basic protocol used - simple mail transport protocol(RFC 821 and 1123). SMTP deals mostly with simple email without attachments. Attachment emails(plus email delivery using some advanced features) uses ESMTP - extended/enhanced SMTP

Very simple protocol,text based, can send email using telnet, very easy to forge an email

telnet grussell.org 25
Connected to grussell.org (10.10.5.5).
Escape character is '^]'.
220 grussell.org ESMTP Sendmail 8.12.11/8.12.11; Sun, 14
Nov 200819:01:01 GMT
> helo pc236b.napier.ac.uk
250 grussell.org Hello pc236b.napier.ac.uk [10.4.5.6],
pleased to meet you
> mail from: g.russell@napier.ac.uk
250 2.1.0 g.russell@napier.ac.uk … Sender ok
> rcpt to: me@grussell.org
250 2.1.5 me@grussell.org.. Recipient ok

> data
354 Enter mail, end with “.” on a line by itself

> From: “Santa" claws@northpole.com
> To: Gordon Russell <me@grussell.org>
> Date: Tue, 15 Jan 2008 16:02:43 -0500
> Subject: SMTP
Hello gordon.
I am Santa
> .
250 Ok: queued as 5555
> QUIT
221 Bye

Envelope and headers
Email information about from and to supplied using SMTP(except the contents of the data command) makes up[what is known as the message envelope, the from,to and other inital information in the email itself(the part in DATA) is known as the email header. The email instruction in the envelope DOES NOT have to match that in the email headers. This is useful when, for intance, dealilng with email mailing lists: the envolope directs it to you, but the headers state it is actually to “the name of the email group”.

Forged Emails
Note that the envelope FROM was: > mail from: g.russell@napier.ac.uk • The data FROM was: From: “Santa" claws@northpole.com • This is perfectly valid. It will be delivered. • The envelope is used through the delivery process, but it is discarded when it is finally delivered to the recipient. • The final recipient cannot recover the information in the envelope. However, the headers can give useful information.

From g.russell@napier.ac.uk Sun Nov 15 11:12:21 2009 Received: from pc236b.napier.ac.uk [10.2.4.5] by grussell.org (8.18.11) id PDQ666 Sun Nov 15 11:12:20 2008 -0000 Received: (gor@localhost) by pc236b.napier.ac.uk (8.18.11) id LXY123 Sun Nov 15 11:12:16 2008 -0000 Date: Sun Nov 15 11:12:15 2008 -0000 From: g.russell@napier.ac.uk To: me@grussell.org Message-Id: <20041115111215.LXY123@pc236b.napier.ac.uk> Subject: Wow Message body is here. This is the message.

From “Date” down is the data added by the orgional sender. As the email moves from machine to machine, extra information is added to the data.

From g.russell@napier.ac.uk Sun Nov 15 11:12:21 2009 Received: from pc236b.napier.ac.uk [10.2.4.5] by grussell.org (8.18.11) id PDQ666 Sun Nov 15 11:12:20 2008 -0000 Received: (gor@localhost) by pc236b.napier.ac.uk (8.18.11) id LXY123 Sun Nov 15 11:12:16 2008 -0000

Received
Received: from pc236b.napier.ac.uk [10.2.4.5] by grussell.org (8.18.11) id PDQ666 Sun Nov 15 11:12:20 2008 -0000 Received: (gor@localhost) by pc236b.napier.ac.uk (8.18.11) id LXY123 Sun Nov 15 11:12:16 2008 -0000
The first “hop” the email went through was at pc236b, email was written by someone on that machine user gor, sendmail handled that hop version 8.18.11

Received: from pc236b.napier.ac.uk [10.2.4.5] by grussell.org (8.18.11) id PDQ666 Sun Nov 15 11:12:20 2008 -0000 Received: (gor@localhost) by pc236b.napier.ac.uk (8.18.11) id LXY123 Sun Nov 15 11:12:16 2008 -0000 • The second “hop” the email went through was at grussell.org. It received the email from a server with ip 10.2.4.5. • Pc236b and 10.2.4.5 should be the same thing. • It took 4 seconds to be delivered between servers. • Sendmail handled that hop, version 8.18.11

Spotting forged emails
Looking for funnies in the headers, dates and times that go backwards(taking into account the timezone), hops which dont match up, strange strings in the hop data, DATA and hop data which makes no sense, HOP routes which sound strange(like a bank delivering emails via yahoo

MX Records
When you email linuxzoo.net, the delivery process will look for a mx record for llinuxzoo.net. If it doesnt find one, email is directed to the A record, if it finds a mx record,email is delivered to the machine described in the MX record. Allows a whole domain to delegate email reception to one or more key servers, without having to have email servers on every single possible host.

Lilnux email
MUA,MTA, and MDA
Email in linux is controlled via three types of servicces:
1. MUA – Mail User Agent
2. MTA – Mail Transfer Agent
3. MDA – Mail Delivery Agent

Mail User Agent(MUA)
The email “client”, users use the mail user agent to read and send emails, it takes email messages which have been delivered to a particular users mailbox and displays them to the user. It takes new messages and passes these to the MMail transfer agent for delivery. Examples include mutt,mail and pine

Mail Transfer Agent
Equivalent to an ip router. Takes messages given by an MUA or another MTA, and depending on the delivery address passes them onto another MTA or to an MDA for delivery. Examples include sendmail,qmail and postfix. Each MTA hop inserts its own data at the start of the email data section.

Mail Delivery Agent
Takes email messages from the MTA and delivers it to a particular user or to a MTA. Once delivered it is held until an MUA for that user reads the email. Examples include mail and procmail for local delivery, and sendmail itself for network delivery.

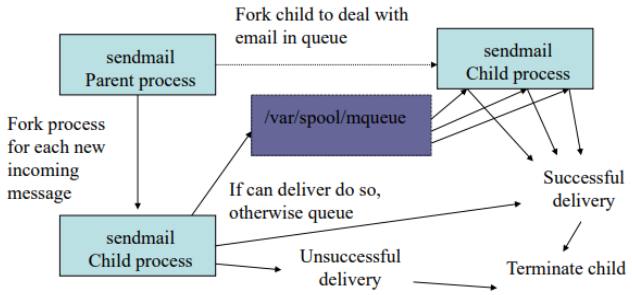
Example • Lets consider an example of g.russell@napier.ac.uk delivering an email to me@grussell.org.

1. g.russell MUA on pc236b.napier.ac.uk send the email to the MTA (sendmail) on localhost. 2. The localhost MTA looks up the MX record for grussell.org. 3. The record indicates that email is delivered to grussell.org itself. 4. Sendmail uses its MDA (SMTP) agent to deliver the email to the MTA on grussell.org. 5. The MTA on grussell.org looks at the destination user (me) and decides that this is a local user. 6. It uses its local delivery agent MDA (procmail) to put this email into the user's mailbox. 7. The email is stored in /var/spool/mail/me. 8. Next time "me" logs into grussell.org, the email will be waiting for him.

Question 1
With respect to email delivery, discuss the difference between a
header and an envelope.
Envelope has the ultimate destination.
Headers contain how the route has been delivered.
Question 2
• When examining an email header, what header elements could be
useful in deciding if it is a forged email. Briefly explain your
Reasoning.
Time zone issues, time going backwards,strange hops that dont make sense,subject might be dodgy

Email Management - Linux
Sendmail, is the oldest most common MTA in use today, many features which are now redundant(like uysing uucp to forward email to the destination using!) huge and prone to hack attack, works well,understood, universally discussed and is still popular.

Sendmail mqueue processing



Aliases
Link recipient envelope address to a local user or action. Sendmail, this is /etc/aliases, file has 1 alias per line, with the alias name, then a :, then action or user.
postmaster : root
me : gor
olduser : /dev/null
automail : | /home/gordon/bin/autoregister.pl
devel : gor, me@grussell.org.uk, a.cumming
Aliases for users, programss(with |) to files (with /) or muliple users(separated withg ,).

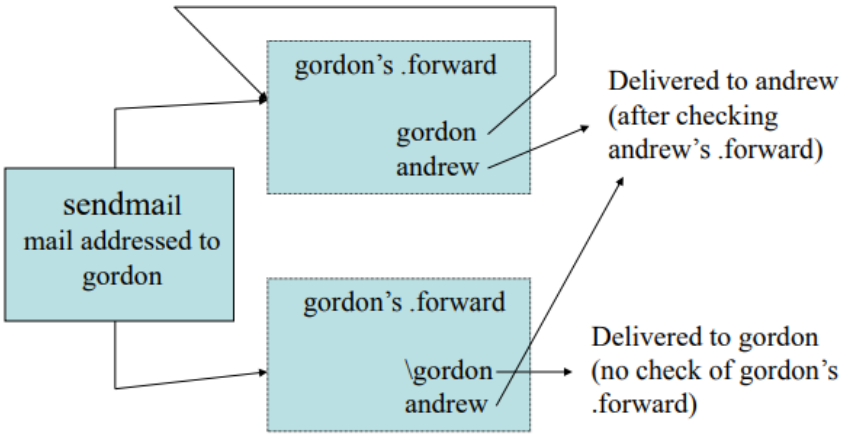
Compiling alias changes

Sendmail does not use /etc/aliases directly, isntead is uses a binary hashed version of the file. This is aliases.db or aliases.dbm. When you change /etc/aliases you must run newaliases to build the hashjed file.

.forward
Only root can make changes to /etc/aliases. Individual users can have simple aliases using .forward in there home directory. Every line in this file is treated as an alias for the user. In this case email for trhis user would go to both grussell and andrew.

Recursive loops
> cat /home/gordon/.forward
gordon
andrew

This appears to suggest delivery as normal to gordon, but also a copy of the email to andrew. • However, for each name the whole alias process is redone, so that (for example) aliases for andrew will also be processed. • Unfortunately “gordon” will trigger access to his .forward file, find “gordon” again, and again, and again. This would break!
• To tell sendmail not to keep looking things up when they are found in .forward, put a \ character in front of the name. In this example: \gordon andrew • This means “no more aliases on gordon”



Sendmail configuration
Config files are in /etc/mail
Mix of .cf and .mc files. .cf is quickly parsed by sendmail but for mortals it is hard to ujndjerstand(although it is flexible). .mc are MACRO level commands which can be sued to build (most) .cf files, stick to .mc for all sensible cases. Run “make”in this directory to convert from .mc to .cf

Basic structure of sendmail.mc
OSTYPE()
Define() - 2 parameters:variable name and value.
define('PROCMAIL_MAILER_PATH','/usr/bin/procmail')
FEATURE()
MAILER() (for example MAILER(SMTP) or MAILER(procmail)
Dnl - a convenience - like a shell #comment

What the options are
Sendmail config is complex, gordens is more than 1000 pages long, basic are straightforward. Most things you want are either features,defines or have their own function name. One looks up the effect they want to have in the documentation, and folows the instructions. Only really keen people would tryt to learn sendmail completely.

Key Issues:Masquerade
Different types of masquerading in sendmail, masquerading is where the identitiy of a machine is disguised for some reason. For instance, mail server is called mail.grussell.org, in a normal setup all your email would appear to come from mail.grussell.org and not just grussell.org.

MASQUERADE_AS('grussell.org')
Now all outgoing emails are rewritten so that the from address appears as grussell.org. It is vital that grussell.org is an A record and not a CNAME.

Relaying
Relaying is the process of asking one mail server to take your email for delivery somewhere else. Can be used by spammers to mass mail junk, but leave hte blame on hte mail server used. It was once switched on by default, which lead to many mail hosts becoming blacklisted. /etc/mail/relay-domains contains a list of domains which this server WILL relay.

Virutal user table and genericstable
Virtual Host - receiving
FEATURE(virtualusertable) supports the /etc/mail/virtusertable file. Sent emails which match the first column are delivered to that specified in the second column.
me@grussell.org gordon
me@grussell.org.uk gordon
jim@grussell.org error:Sorry he has left
@grussell.org gradmin

Virtual Hosts - sending
FEATURE(genericstable) and /etc/mail/genericstable does virtual host mapping for outgoing mail.
gordon me@grussell.org
andrew andrew@grussell.org.uk
dbuser gordon@db.grussell.org

Access
FEATURE(access_db) and /etc/mail/access allows hosts and domains to be controlled on a per-user basis
lycos.com DISCARD
grussell.org.uk RELAY
gordon@sqlzoo.net RELAY
sqlzoo.net ERROR:5.0.0:550 No way

Possible responses.
• OK – local recipients only

• RELAY – forward
• REJECT – no way!
• DISCARD – no way quietly
• ERROR – Send an error back to the sender (RFC 1893 compliant codes).

Are a number of black hole email lists. Lists populated by suspected spammer Such lists include: – http://www.spamhaus.org/
So to drop people in the sbl list do:
FEATURE(`dnsbl', `sbl.spamhaus.org', `"550 Mail from " $`'&{client_addr} " refused - see http://www.spamhaus.org/SBL/"')dnl

Spamhatus
Spamhatus actually maintain 3 lists:– SBL – List of spammers and the like – XBL – Machines which seem to be exploited in some way. For instance, have open proxies, or perhaps a worm or virus. – PBL – A list of machines which should not really be sending email. For instance, the dynamic ips of customers in an ISP, where they all should be using the ISPs SMTP server for sending email.List are only as good as the data used to build them. However, good lists can generally be trusted, so long as you fully understand the implications of using them.

Fighting SPAM
Few research areas where investigators are trying to fight spam. One tech is called SPF - sender policy framework. When a email is recieved suggesting it is from linuxzoo.net the MTA does a dns lookup on linuxzoo.net, looking for a TXT record. If it finds one starting with v=spf1, then some additional checkls are performed before the email is processed

SPF for linuxzoo
Txt record supporting spf in linuxzoo is:
v=spf1 ip4:146.176.166.1 ip4:146.176.166.15 a ~all
This indicates that email can only be send by 146.176.166.1, 146.176.166.15, or from an IP which matches the A record for linuxzoo.net. • If any of those things are true, the mail is processed normally. If it is all false, the MTA carries out the “~” action.

Actions
Really it should ahve “-all” indicates that a failed test result in the email being rejected. “~all” is a soft fail, really for debugging, indicates that it is probably junk but not confident enough to guarntee it. Until then it means that if the SPF rules are borken, then delete the email if you are being tough, in reality it probably means take no action

Commaands that werent in the lecture
file filename
Says what kind of file this is based on the signature that is contained in it